OPTIMUM®
MASCHINEN - GERMANY

Computer Numeric Control

- CNC milling machines
- CNC lathes
- CNC software
- Accessories

CNC metal working machines

The OPTIMUM in quality, price-performance and service



OPTI mill®

OPTI turn®









CNC main catalogue 2013/14



Headquarters in Hallstadt/Bamberg





Assembly F80



Assembly CNC changer



Final inspection L 33



Manufacturing on CNC machines

Quality at a cost-efficient price
OPTIMUM metal working machines

Your requirements are our target

Since more than 20 years we are dealing with the development, design and production of **OPTIMUM** machines. All our products distinguish themselves by quality, accuracy, sustainability and consistent value. In addition to our production plants we have our articles solely produced by manufacturers who are able to fulfil our high quality requirements. The knowledge of our experience of decades serves to constantly advance and improve our products.

The products

You're looking for a functional CNC machine offering comprehensive features at an economical price? Then you make the right choice by purchasing an **OPTIMUM** metal working machine. Our products are convincing by outstanding quality, accurate manufacturing and offer an “OPTIMUM” in price and performance.

Production

Since 2003, **OPTIMUM** produces a large part of its metal working machines in its own factory in Yangzhou in China. The quality is monitored by German quality management representatives. Another important part of the **OPTIMUM** metal working machines is exclusively produced or produced for **OPTIMUM** according to special specifications.

Quality

The **OPTIMUM** products are produced with high quality requirements. There are the same or similar products on the market which by far do not attain the quality standard of **OPTIMUM**: **OPTIMUM** products are **often copied.**



Optimum factory, Yangzhou China

The OPTIMUM guarantee of quality
All OPTIMUM products are produced with high quality requirements.

Distribution

OPTIMUM CNC machines can be purchased from your competent specialized dealers on site. An area-wide network of specialized dealers and sales partners - who are supported by the employees of **OPTIMUM** are at your disposal to help you with words and deeds.

Exhibition

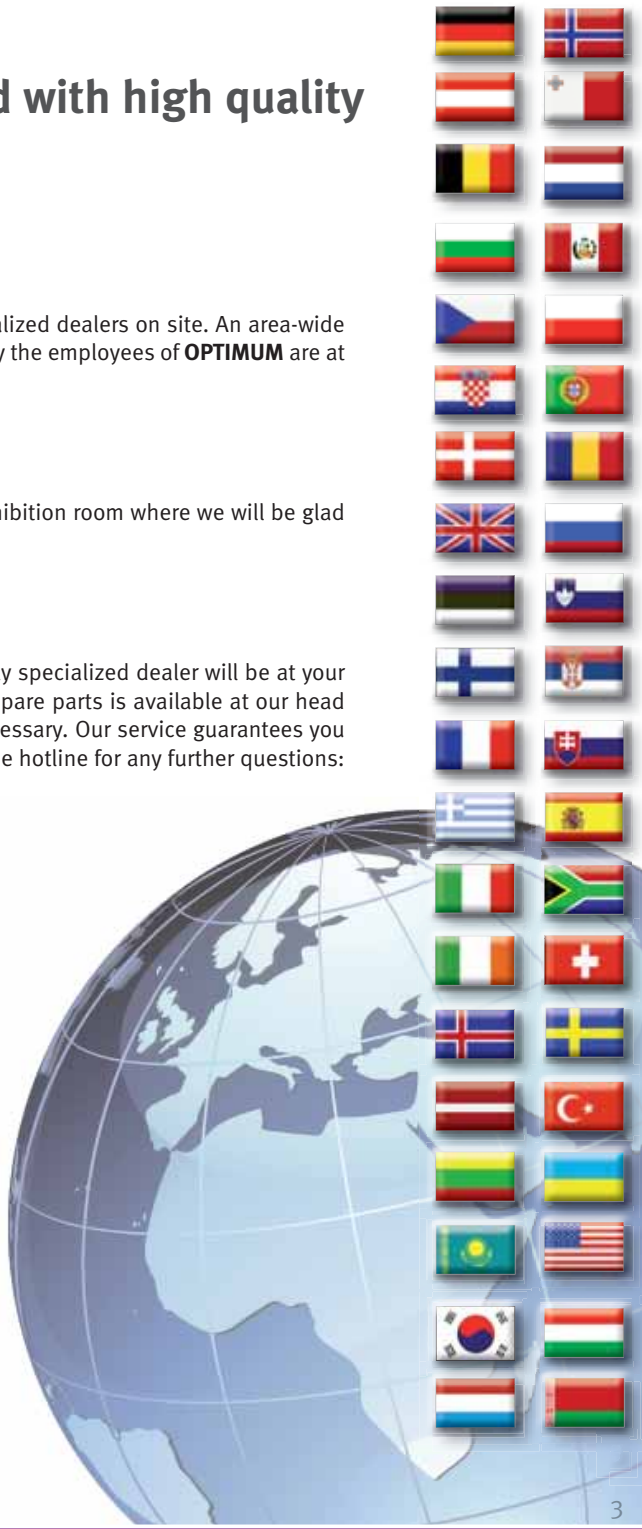
At our site in Hallstadt/Bamberg you will find a largely dimensioned exhibition room where we will be glad to present our products to you.

After-Sales-Service

If your machine requires an after-sales service, we as well as your locally specialized dealer will be at your disposal with an experienced service team. A comprehensive stock of spare parts is available at our head office in order to ensure the supply of spare parts at short notice, if necessary. Our service guarantees you the value of your machines for years. Please use our **free of charge** phone hotline for any further questions: +49/ (0) 951 - 96 555 74

Worldwide representations

As a German company, **OPTIMUM** is globally working together with specialist partners from its company's headquarters at Hallstadt/Bamberg.
The international distribution network of **OPTIMUM** extends worldwide over the most important countries.





Kilian Stürmer
Manager

Optimum Maschinen Germany
New cooperation partner for CNC training in Bavaria





A strong team

Since many years Siemens is the system supplier of the control and drive engineering for the CNC controlled lathes and milling machines of the Optimum Maschinen Germany GmbH. Due to the long term and successful cooperation, a cooperation partnership for the CNC training was agreed in June 2012.
In target group specific courses the participants of the training program are made familiar with the functions of the different Sinumerik controls.
To do so, Optimum uses CNC machines with the Sinumerik controls 828D, 802S and 808D. Also the software "Sinumerik Operate" with the working step programming "ShopMill" and "ShopTurn" are part of the training courses. The machine operators become familiar with the rapid NC programming, the work preparation and the intuitive use of the software. Trainers of OPTIMUM certified by Siemens train the skilled workers either directly at their working places or at the company headquarters of Optimum at Hallstadt nearby Bamberg

With knowledge to success!

You may learn which inumerous application options are offered by programming with different software programs and machine controls in the field of CNC engineering.

Operation and programming using
Siemens SINUMERIK 828D
Siemens SINUMERIK 802S
Siemens SINUMERIK 828D Operate ShopMill, ShopTurn
Siemens SINUMERIK 808D

Training services
Perfectly qualified specialized staff is the prerequisite for productivity and quality. The SINUMERIK training of OPTIMUM establishes this basis. In target group specific courses the participants are made familiar with the functions of the SINUMERIK CNC controls.





Trainers certified by Siemens will train you on site in your premises worldwide or at our company headquarters in Hallstadt/Bamberg.
Please do not hesitate to contact us: info@optimum-maschinen.de



Sales training and training of staff / customer training

The success in handling our products is mainly determined by passing the technical knowledge on to the sales and service employees by consequent internal training measures. This qualification forms the basis so that we are able to guarantee a skilled, qualified consultation and troubleshooting at any time when a customer contacts us. This qualification also includes our customers; we are providing workshops at our facility or training courses at the customer's premises.



Service support

...since good service is essential!
Our OPTIMUM technicians are at your disposal at any time in case of service in order to support the shop in performing a rapid and targeted repair.
Weak points are analysed and customer wishes noted in close cooperation with the service centre and its internal and external employees. These points are evaluated under technical points of view and noted down in specific requirements or measure catalogues.



Testing in practice

Our engineers combine theory and practice. In order to leave nothing to chance with regard to the product satisfaction; application-specific tests are performed on all machines and selected customers are consequently involved in this testing. In this way, any new product first has to prove itself during everyday work before it is included as a inherent part in our range of products. The technical analysis helps us to recognize and resolve any remaining weak points.



Technical customer support

Our customers may rightly expect that our specialists are using their knowledge and experience to their utmost satisfaction. Our product consultants support the user with technical information. A phone hotline as well as an info mail account are available which our customers may use in order to receive qualified information or troubleshooting at short notice.



Planning

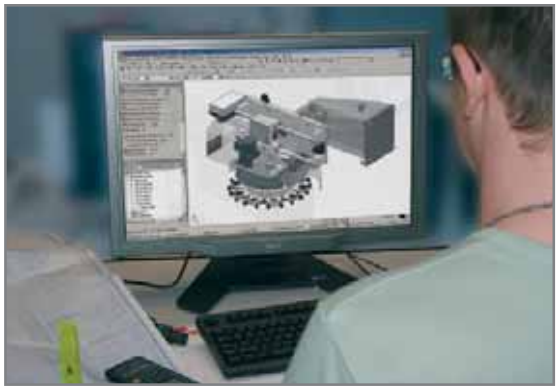
Already during the planning phase our technical department intervenes in the development of new products which are manufactured at our facility as well as at the facilities of our partners. In this way, all marketing influences and customer requirements are immediately added where the future direction for a successful product design is set.

Our technical department has a well-established team of technically highly qualified specialists. Their well-founded expertise allows flexible and creative implementation of all requirements which are demanded for our product and services.



Technical documentation and risk analyses

Our technical authors are able to achieve a high standardized level which complies with all legal requirements or even exceeds them. These enormous efforts only serve to easily introduce the machine to our customers as well as to guarantee the continuous and safe operation.
In order to compensate the identified safety gaps, measures are taken to minimize the risk in the frame of the analysis. Then an evaluation of the remaining residual risk is performed after having implemented these measures.



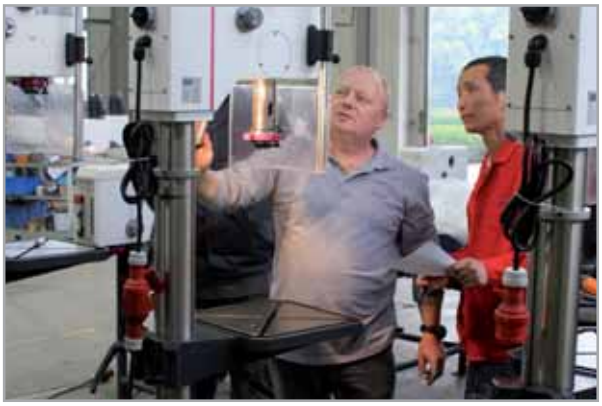
Development

The development work is performed by creating a virtual machine model using most modern 3D CAD software. In doing so, beside the optimum functionality of the machine we also pay attention to the re-usability of the data which are created during the development phase. Such data do not only serve to elaborate manufacturing documents and documentations, but also to establish calculations as well as for computer-aided manufacturing and to visualize and simulate the product.



Quality management incoming goods inspection

Our quality manager from Germany is the first contact for adhering to quality on site.
He is responsible for the dimensional accuracy of the components as well as their control and the quality of the manufacturing process. He is closely working together with our technical department in the headquarters in Germany and thus forming an optimum symbiosis.



Quality management production and outgoing goods inspection

Beside delivery reliability and service the quality of our products is of highest importance for us. Our quality is ensured by continuous internal and external monitoring on site by our quality manager.

Our comprehensive outgoing goods inspection is performed according to the general technical guidelines.



Supply of replacement parts

We know that the rapid availability of spare parts is one of the basic requirements for a satisfying, customer-focused service concept.
At Optimum Machines Germany GmbH the planning, coordination and availability of replacement parts are harmonized in a way to ensure maximum benefit regarding economic efficiency and speed.

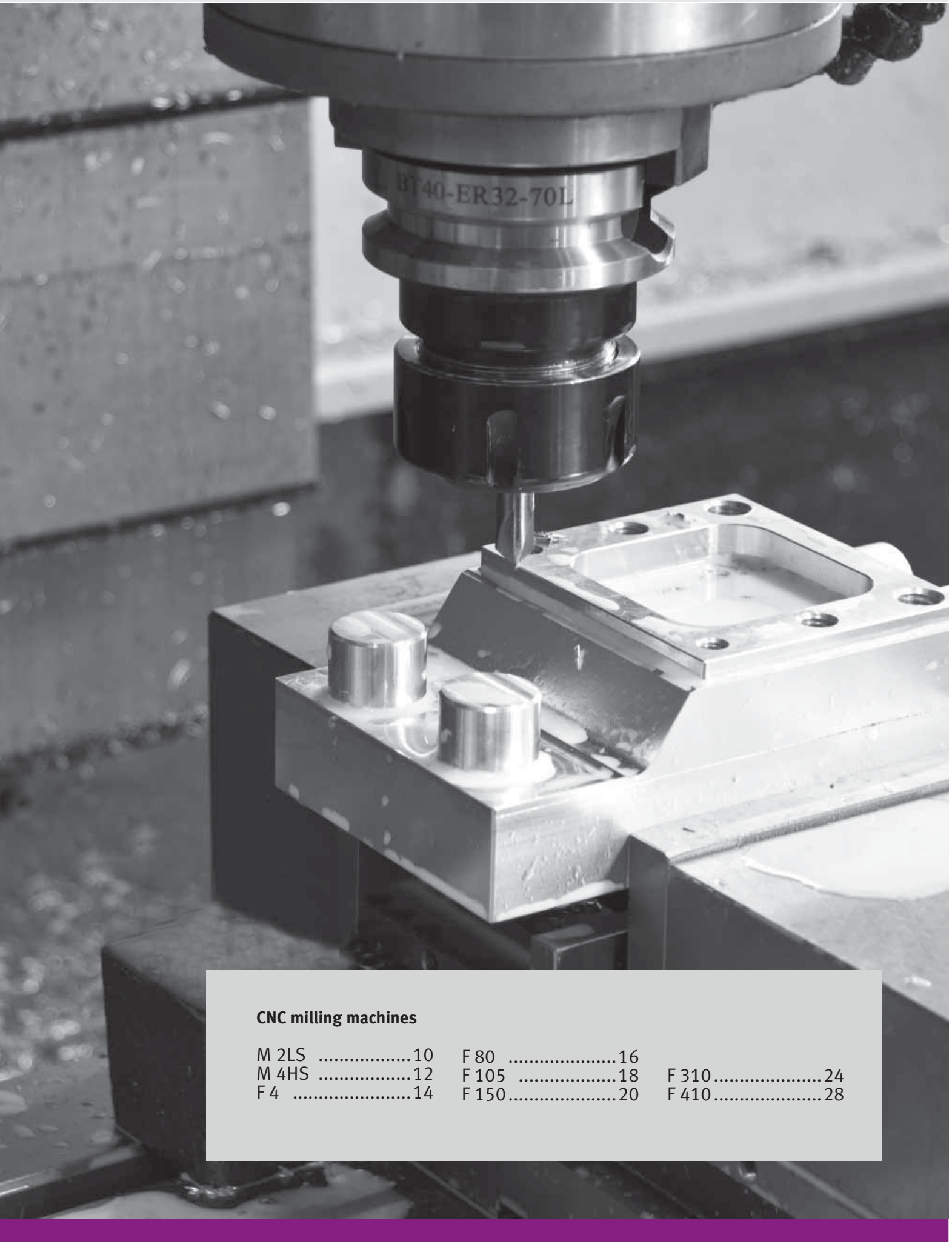
OPTImill®

OPTIturn®

Two-stage price-performance ratio

We subdivide our metal working machines into two categories with an "OPTIMUM" in quality and price-performance.

Standard line	- Metal working machines for the professional craftsmen and industrial enterprises.
Premium line	- Machines for small and medium-sized companies and for the industry



CNC milling machines

M 2LS 10
M 4HS 12
F 4 14
F 80 16
F 105 18
F 150 20
F 310 24
F 410 28



CNC milling machines

NEW

CNC milling machine with servo-drives and SIEMENS control SINUMERIK 808 D, perfect for training purposes

Convincing arguments: quality, efficiency and price

- Ball screw made by HIWIN in all three axes
- Robust and precise dovetail guide with adjustable V-ledges free of play
- Bellow as guide protection
- Electronic controllable drive
- Front door with safety switch
- Base designed on two sides. Front side with access to the coolant tank; rear side with access to the electronic system
- Central lubrication
- Gear stage is represented with the gear change to the SIEMENS control

Servo-drives
- Higher speed
- Better accuracy

2 gear stages (L/H)
- Powerful transmission of the motor power

Signal lamp

SIEMENS control
- in a slewable housing

Working area
- Can be easily viewed from three sides
- Clear, resistance Makrolon panes



Handwheel
- Electronic system
- Reduces the setting time to a minimum
- Emergency-stop push button
- Acknowledgement button



Housing design
- Compact and functional



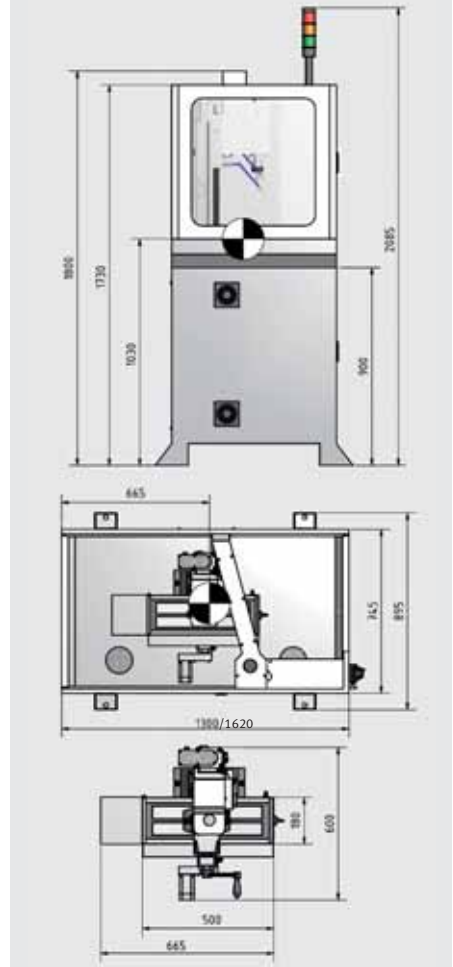
„CNC-instruction - Information“ on page 80

Coolant
- Tank integrated in the base
- Tank capacity 10 litres



Type	M2LS
Item No	350 0123
Electrical connection	
Connection ~50 Hz	230 V / 1 Ph
Total connected value	1.0 kW
Driving motor	850 W
Motor coolant pump	40 W
Tank capacity coolant pump	10 litres
Spindle seat	
Spindle seat	MT 2/M 10
Milling head dimension max.	Ø 52 mm
End mill dimension max.	Ø 16 mm
Travels	
X axis	480 mm
Y axis	155 mm
Z axis	280 mm
Feed speeds	
X axis	2'000 mm/min.
Y axis	2'000 mm/min.
Z axis	2'000 mm/min.
Throat	
Throat	185 mm
Speeds	
Gear stages slow	90 - 1'480 min^{-1}
Gear stages rapid	150 - 3'000 min^{-1}
Gear stages	2 steps, infinitely variable
Cross table	
Table length x width	700 x 180 mm
T-slot size / number / distance	12 mm / 3 / 63 mm
Load (max.)	55 kg
Dimensions	
Length	1'620 mm
Width x Height	895 x 2'085 mm
Total weight	380 kg

Dimensions





Standard equipment

Operating tool
Data cable RS 232
Siemens Tool Box CD
Working lamp
Taper mandrel MT 2, B16 M10

Siemens SINUMERIK 808 D

Outstanding performance. Simply intelligent.

- User interface Sinumerik Operate Basic with Sinumerik startGuide for a new dialogue-oriented user support which conducts through the setting of the machine step-by-step and reduces the time for commissioning to a minimum.
- 7.5“ LCD colour display
- Sinumerik MDynamics (Look ahead, Advanced Surface ...)
- High-speed settings Cycle (Cycle 832)
- Look Ahead kits 50
- 80 bit Nano accuracy
- DIN / ISO programming
- Contour calculator
- Online help
- Max. 3 interpolating axes
- Memory size 1.25 MB (extendable via USB
- Metric/Inch change-over
- ProgramGUIDE BASIC



Stable CNC milling machine with SIEMENS - control SINUMERIK 802S base line, perfect for the users who need high speeds and high accuracy

Convincing arguments: quality, efficiency and price

- Ball screw made by *HIWIN* in all three axes
- Solid, precise cross table, largely dimensioned and precisely machined surface
- Heavy, solid cast iron construction
- Re-adjustable V-ledges of the X and Y axis
- Base designed on two sides. Front side with access to the coolant tank; rear side with access to the electronic system and to the Optimum CNC Controller IV
- Infinitely variable drive
- SIEMENS Micromaster as frequency converter
- Front doors with safety switch
- Central lubrication
- Linear guideway in the Z-axis
- Very wide speed range
- Powerful micro processor control
- Control of the axis drives
- Micro step setting is possible
- Spindle seat ER32 for high speed and normal chipping

Housing design
- Compact and functional
- Lateral parts of housing are collapsible thus easy access to the machine

Working area
- Easily viewed from three sides
- Clear, resistance Makrolon panes

- **Signal lamp**

SIEMENS control
- in slewable housing

Handwheel
- Electronic system
- Reduces the setting time to a minimum
- Emergency-stop push button
- Acknowledgement button



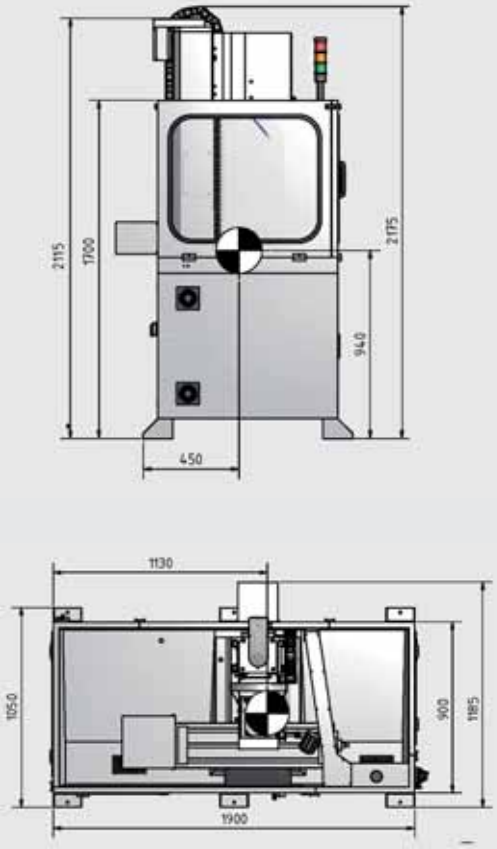
Fig.: M 4 HS



Rear of base
- Powerful micro processor control
- Control of the axis drives
- Micro step setting is possible

Type	M 4HS
Item No	350 0140
Electrical connection	
Connection ~50 Hz	400 V / 3 Ph
Total connected value	3.1 kW
Driving motor	3 kW
Torque driving motor	4.7 Nm
Gear stages driving motor	Direct transmissions
Motor coolant pump	40 W
Tank capacity coolant	15 litres
Spindle seat	
Spindle seat	ER 32
Drilling capacity in steel (S235JR)	Ø 13 mm
Continuous drilling capacity in steel (S235JR)	Ø 11 mm
End mill dimension max.	Ø 20 mm
Travels	
X axis	500 mm
Y axis	240 mm
Z axis	530 mm
Feed speeds (were determined in the 1/16 step procedure)	
X axis	2'000 mm/min.
Y axis	2'000 mm/min.
Z axis	2'000 mm/min.
Throat	
Throat	260 mm
Speeds	
Speeds	150 - 8'600 min^{-1}
Cross table	
Table length x width	750 x 210 mm
T-slot size / number / distance	18 mm / 3 / 80 mm
Dimensions	
Length x Width x Height	1'900 x 1'185 x 2'175 mm
Total weight	800 kg

Dimensions



Standard equipment

- Operating tool
- Data cable RS 232
- Siemens Tool Box CD
- Working lamp
- Coolant equipment

- We imperatively recommend the briefing performed by our specialists.



„CNC-instruction - Information“ on page 80

Siemens 802S base line



SINUMERIK 802S base line, the CNC control made by SIEMENS for application in the low power range is particularly well suited for milling machines with up to 3 axis and a spindle and distinguishes itself by the following properties:

- Extremely easy operation
- Maintenance-free
- Compact design.

Milling cycles:

- Deep-hole drilling
- Row of holes - circle
- Row of holes - line
- Drilling, countersinking
- Milling of pockets, grooves and circular pockets

NEW

Stable CNC milling machine with SIEMENS control SINUMERIK 808 D

Convincing arguments: quality, efficiency and price

- Linear guideways in all axis
- All axis equipped with ball screws
- 8-position magazine tool changer
- Cross table with 3 T-slots
- Electronic handwheel
- Coolant equipment
- Central lubrication
- Servo drive in all axis
- Slewable control panel
- Siemens Sinumerik 808 D on PC free of charge (Computer workstation)
- Including 2-year Siemens warranty

machine in action



Cable duct
- in an energy chain

SIEMENS SINUMERIK 808D

SIEMENS control
- in a slewable housing

Working area
- Can be easily viewed from three sides
- Clear, resistance Makrolon panes

- **Signal lamp**



Connections
- Easily accessible
- Easy to connect to the machine
- Jack
- USB
- RS232

Handwheel
- Electronic system
- Reduces the setting time to a minimum
- Emergency-stop push button
- Acknowledgement button

„Starter kit BT 30“ on page 76



„CNC-instruction - Information“ on page 80

Convincing arguments ...



Tool change system
- Automatic by means of magazine disc for 8 tools
- **Tool replacement speed 8 sec.**
- Max. tool diameter 50 mm



Control cabinet
- Clearly arranged structure
- Largely dimensioned



Central lubrication
- reduced wear and increases the service life of the machine

Power connection
- Quick action coupling



Chip tray
- Simple chip disposal due to extractable chip tray (backside)

Type	F 4
Item No	350 0430
Electrical connection	
Connection ~50 Hz	400 V / 3 Ph
Total connected value	8.5 kW
Driving motor	2.2 kW
Torque driving motor	14 Nm
Motor coolant pump	95 W
Tank capacity coolant	
Spindle seat	
Spindle seat	ISO 30/BT 30
Milling head dimension max.	Ø 65 mm
End mill dimension max.	Ø 30 mm
Milling accuracy	
Repeatability	± 0.015 mm
Positioning accuracy	± 0.015 mm
Tool changer	
Disc type	8 tools
Tool diameter max.	70 mm
Tool weight max.	6 kg
Time - tool changing	8 seconds
Travels	
X axis	310 mm
Y axis	200 mm
Z axis	300 mm
Feed speeds	
Rapid feed X-, Y-, Z axis	10'000 mm/min.
Torque motor	
X axis	3.3 Nm
Y axis	5 Nm
Z axis	5 Nm
Speeds	
Speeds	50 - 9'000 min^{-1}
Cross table	
Distance spindle - table	90 - 390 mm
Table length x width	690 x 210 mm
T-slot size / number / distance	16 mm / 3 / 63 mm
Load (max.)	100 kg
Air connection	
Connection	min. 7 bars
Dimensions	
Length x Width x Height	1'800 x 1'641 x 2'000 mm
Tank capacity coolant	50 litres
Total weight	1'800 kg

Siemens SINUMERIK 808 D

Outstanding performance. Simply intelligent.

- User interface Sinumerik Operate Basic with Sinumerik startGuide for a new dialogue-oriented user support which conducts through the setting of the machine step-by-step and reduces the time for commissioning to a minimum.
- 7.5“ LCD colour display
- Sinumerik MDynamics (Look ahead, Advanced Surface ...)
- High-speed settings Cycle (Cycle 832)
- Look Ahead kits 50
- 80 bit Nano accuracy
- DIN / ISO programming
- Contour calculator
- Online help
- Max. 3 interpolating axes
- Memory size 1.25 MB (extendable via USB
- Metric/Inch change-over
- ProgramGUIDE BASIC



NEW

The compact solution in the field of training with SIEMENS control 808D

Convincing arguments: quality, efficiency and price

- Main spindle drive with servo motor
- Spindle motor with 8'000 min^{-1} as standard
- All linear guiding are equipped with special steel covers
- Automatic central lubrication of the machine
- Integrated machine lamp / integrated coolant equipment
- Tool change is performed automatically or by pushing a button (electro-pneumatic tool clamping device)
- Solid, precise cross table, largely dimensioned and precisely machined surface
- Largely designed access doors in order to reduce cleaning and maintenance times to a minimum
- Electronic handwheel reduces the set-up time to a minimum with EMERGENCY-STOP push button and acknowledgement button
- Servo drive in all axis
- Including 2-year Siemens warranty
- Siemens Sinumerik 808 D on PC free of charge (Computer workstation)



„CNC-instruction - Information“ on page 80

„Starter kit BT 30“ on page 76

Convincing arguments ...



Tool change system
- Automatic by means of magazine disc for 10 tools
- Tool replacement speed 7 sec.
- Tool diameter max. 104 mm



Control cabinet
- Clearly arranged structure
- Largely dimensioned



LED-Machine lamp
- High light output
- Little energy consumption
- Long service life



Cable duct
- in an energy chain

Type	F 80
Item No	350 1080
Electrical connection	
Connection ~50 Hz	400 V / 3 Ph
Total connected value	9.5 kW
Driving motor	2.2 kW
Torque driving motor	14 Nm
Motor coolant pump	650 W
Tank capacity coolant	
Spindle seat	
Spindle seat	ISO 30 BT30
Milling head dimension max.	Ø 65 mm
End mill dimension max.	Ø 30 mm
Milling accuracy	
Repeatability	± 0.015 mm
Positioning accuracy	± 0.015 mm
Tool changer	
Disc type	10 tools
Tool diameter max.	104 mm
Tool weight max.	6 kg
Time - tool changing	7 seconds
Travels	
X axis	400 mm
Y axis	225 mm
Z axis	375 mm
Feed speeds	
Rapid feed (X, Y, Z axis)	10'000 mm/min.
Torque motor	
X axis	5 Nm
Y axis	5 Nm
Z axis	6 Nm
Speeds	
Speeds	50 - 8'000 min^{-1}
Cross table	
Distance spindle - table	75 - 475 mm
Table length x width	800 x 260 mm
T-slot size / number / distance	16 mm / 5 / 50 mm
Load (max.)	150 kg
Dimensions	
Length x Width x Height	1'925 x 1'600 x 2'070 mm
Tank capacity coolant	160 litres
Total weight	1'900 kg

Siemens SINUMERIK 808 D

Outstanding performance. Simply intelligent.
- User interface Sinumerik Operate Basic with Sinumerik startGuide for a new dialogue-oriented user support which conducts through the setting of the machine step-by-step and reduces the time for commissioning to a minimum.
- 7.5“ LCD colour display
- Sinumerik MDynamics (Look ahead, Advanced Surface ...)
- High-speed settings Cycle (Cycle 832)
- Look Ahead kits 50
- 80 bit Nano accuracy
- DIN / ISO programming
- Contour calculator
- Online help
- Max. 3 interpolating axes
- Memory size 1.25 MB (extendable via USB
- Metric/Inch change-over
- ProgramGUIDE BASIC



OPTImill® F 105

NEW

The compact solution to produce small series in small and medium-sized companies and in the field of training with SIEMENS control 808D

Convincing arguments: quality, efficiency and price

- Robust type
- Main spindle drive with servo motor
- Spindle motor with 8'000 min^{-1} as standard
- All linear guiding are equipped with special steel covers
- Automatic central lubrication of the machine
- Integrated machine lamp / integrated coolant equipment
- Tool change is performed automatically or by pushing a button (electro-pneumatic tool clamping device)
- Solid, precise cross table, largely dimensioned and precisely machined surface
- Largely designed access doors in order to reduce cleaning and maintenance times to a minimum
- Electronic handwheel reduces the set-up time to a minimum with EMERGENCY-STOP push button and acknowledgement button
- Servo drive in all axis
- Feed motors made by SIEMENS (V60)
- Including 2-year Siemens warranty
- Siemens Sinumerik 808 D on PC free of charge (Computer workstation)



- **Signal lamp**

Handwheel
- Electronic system
- Reduces the setting time to a minimum
- Emergency-stop push button
- Acknowledgement button

Storage shelf
- for tools

„CNC-instruction - Information“ on page 80

„Starter kit BT 40/Set 1“ on page 77
„Starter kit BT 40/Set 2“ on page 78

Convincing arguments ...



Tool change system
- Automatic by means of magazine disc for 12 tools
- Tool replacement speed 7 sec.
- Tool diameter max. 104 mm



Control cabinet
- Clearly arranged structure
- Largely dimensioned



LED-Machine lamp
- High light output
- Little energy consumption
- Long service life



Connection
- Compressed air
- Central lubrication

Type	F 105
Item No	350 1105
Electrical connection	
Connection ~50 Hz	400 V / 3 Ph
Total connected value	10 kW
Driving motor	5.5 kW
Torque driving motor	35 Nm
Motor coolant pump	650 W
Tank capacity coolant	
Spindle seat	
Spindle seat	ISO 40 BT40
Milling head dimension max.	Ø 75 mm
End mill dimension max.	Ø 35 mm
Milling accuracy	
Repeatability	± 0.01 mm
Positioning accuracy	± 0.01 mm
Tool changer	
Disc type	12 tools
Tool diameter max.	104 mm
Tool weight max.	6 kg
Time - tool changing	7 seconds
Travels	
X axis	550 mm
Y axis	305 mm
Z axis	460 mm
Feed speeds	
Rapid feed (X, Y, Z axis)	10'000 mm/min.
Torque motor	
X axis	5 Nm
Y axis	5 Nm
Z axis	10 Nm
Speeds	
Speeds	10 - 8'000 min^{-1}
Cross table	
Distance spindle - table	100 - 600 mm
Table length x width	800 x 320 mm
T-slot size / number / distance	14 mm / 3 / 100 mm
Load (max.)	300 kg
Dimensions	
Length x Width x Height	2'164 x 1'860 x 2'200 mm
Tank capacity coolant	200 litres
Total weight	2'800 kg

Siemens SINUMERIK 808 D

Outstanding performance. Simply intelligent.
- User interface Sinumerik Operate Basic with Sinumerik startGuide for a new dialogue-oriented user support which conducts through the setting of the machine step-by-step and reduces the time for commissioning to a minimum.
- 7.5“ LCD colour display
- Sinumerik MDynamics (Look ahead, Advanced Surface ...)
- High-speed settings Cycle (Cycle 832)
- Look Ahead kits 50
- 80 bit Nano accuracy
- DIN / ISO programming
- Contour calculator
- Online help
- Max. 3 interpolating axes
- Memory size 1.25 MB (extendable via USB
- Metric/Inch change-over
- ProgramGUIDE BASIC



The Optimum Premium CNC milling machine with SIEMENS control SINUMERIK 828 D distinguishes itself by performance, speed, accuracy and high durability

Convincing arguments: quality, efficiency and price

- Heavy type
- High productivity
- Linear guiding for high rapid feed speeds in all axis
- 16-position magazine tool changer (carousel type)
- High reliability
- Cast iron body made of quality Meehanite cast iron
- Cross table precision ground with 4 T-slots
- Heavily ripped, torsion-free machine basis
- Precision ground pre-tensed high-performance ball screws in all axes Ø 32 x P8 x C3
- Powerful servo drives directly on the ball screws in all 3 axis
- Electronic handwheel
- Chip conveyor
- Cleaning gun
- RJ45 plug connection, USB connection and power connection (230V)
- Coolant equipment with coolant tank with a capacity of 210 litres
- Heat exchanger
- Including 2-year Siemens warranty

Including

Our options:

- **Safety Integrated**
- **Residual material detection and operation**
- **ShopMill Working step programming**
- **Managing network drive**
- **3-D simulation finished part**
- **Simultaneous recording**

Siemens Safety Integrated

Set up with opened door

safety integrated power by siemens

SIEMENS SINUMERIK 828D

PREMIUM

Special equiment:
Tool changer 24-position double arm gripper

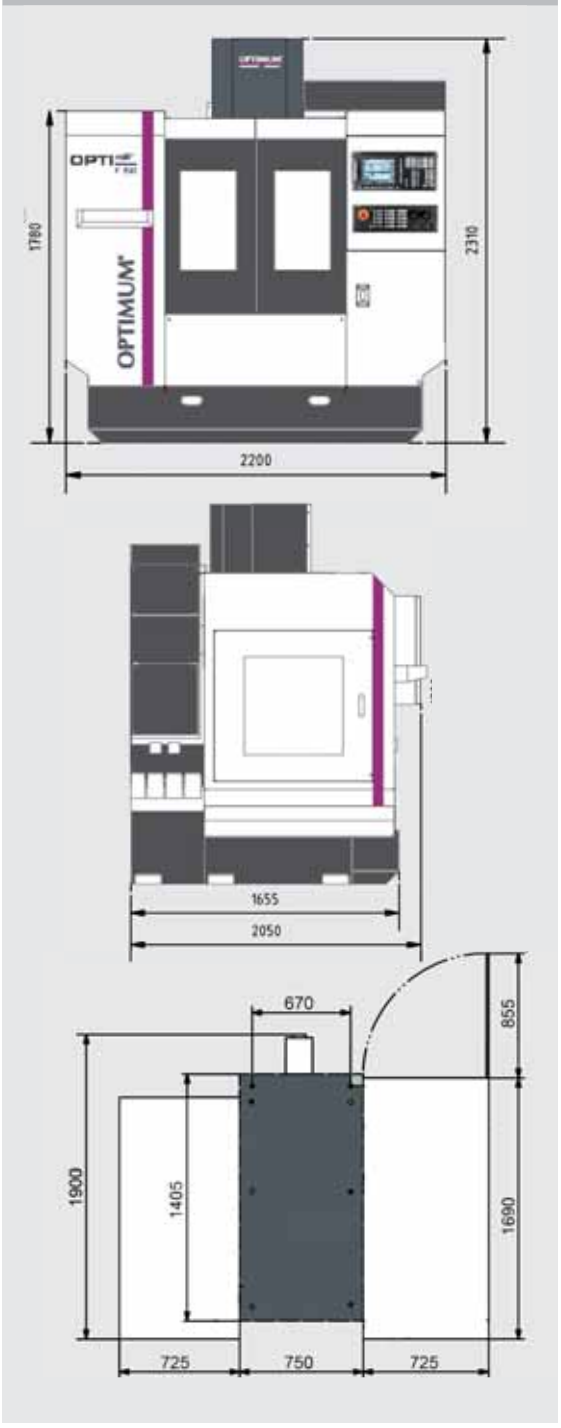
- Time - tool changing Tool to tool 1,8 Sek.



Fig. with optional double arm gripper and chip carriage

Type	F 150
Item No	351 1210
Electrical connection	
Connection ~50 Hz	400 V / 3 Ph
Total connected value	25 kW
Driving motor	12 kW (S1-mode; 9 kW)
Torque driving motor	57 Nm
Motor coolant pumps 3 pcs.	1.27 kW each
Spindle seat	
Spindle seat	ISO 40 DIN 69871
Milling head dimension max.	Ø 63 mm
End mill dimension max.	Ø 32 mm
Milling accuracy	
Repeatability	0.005 mm
Positioning accuracy	± 0.005 mm
Tool changer	
Disc type	16 tools
Tool diameter max.	89 mm
Tool weight max.	8 kg
Time - tool changing Tool to tool	7 seconds
Travels	
X axis	760 mm
Y axis	430 mm
Z axis	460 mm
Feed speeds	
Milling feed (X, Y, Z axis)	10'000 mm/min.
Rapid feed (X-, Y-, Z axis)	24'000 mm/min.
Torque motor	
X axis	6 Nm
Y axis	6 Nm
Z axis	11 Nm
Speeds	
Speeds	10 - 10'000 min^{-1}
Cross table	
Distance spindle - table	102 - 562 mm
Throat	480 mm
Table length x width	900 x 410 mm
T-slot size / number / distance	16 mm / 4 / 102 mm
Load (max.)	350 kg
Dimensions	
Length x Width x Height	3'000 x 2'050 x 2'310 mm
Tank capacity coolant	210 litres
Total weight	3'520 kg

Dimensions



Standard equipment

Residual material detection and machining for contour pockets and stock removal	Coolant pistol
Shop Mill	Chip flushing system
Network drive management	Working lamp
3-D simulation finished part	Tool changer 16-position carousel type
Simultaneous recording	Chip conveyor screw-conveyor type
Safety integrated	Heat exchanger
Sinumerik Operate	6 pcs. Machine feet
EMC	

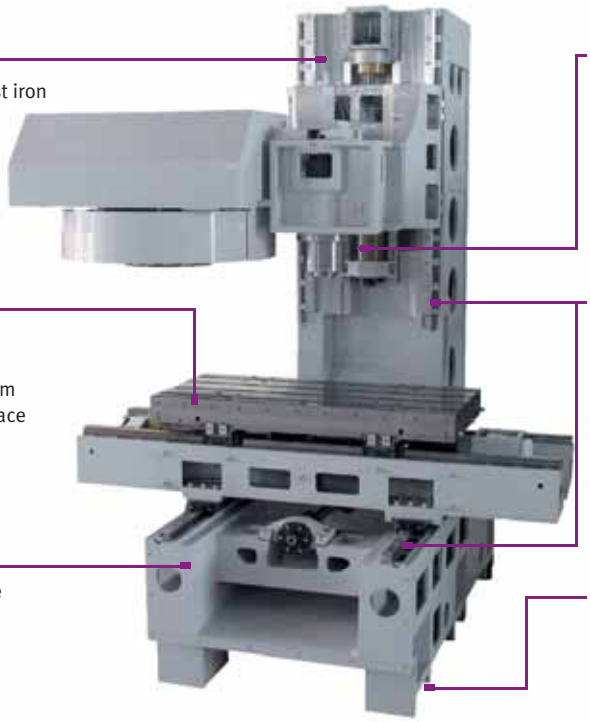
„Starter kit SK40/DIN 69871“ on page 79



„CNC-instruction - Information“ on page 80

Standard equipment

Convincing arguments: quality, efficiency and price



Cast iron body
- made of quality Meehanite cast iron

Spindle
- ISO 40
- Spindle speed 10'000 min^{-1} as standard

Cross table
- Massive, accurate and largely dimensioned
- Clamping surface 900 x 410 mm
- Precisely machined at the surface

Linear guideways
- for high rapid feed speeds 24'000 min^{-1} in all three axes

Base body
- Quality cast iron of ripped type

Machine feet
- 6 pieces
- Optimum alignment of the machine

Heat exchanger



- Efficient integrated heat transfer from the switch cabinet
- Avoids that faults are generated from the overheated electrical and CNC components

Tool change system



- Automatic by means of magazine disc for 16 tools
- Max. tool length 300 mm

Handwheel



- Electronic system
- Reduces the set up time
- Emergency-stop push button
- Acknowledgement button

Cleaning gun



- Easy cleaning of the working area

Chip flushing system



- Powerful chip flush system to clean the working chamber and the workpiece

Laser measurement



- Guaranteed repeatability and positioning accuracy

Special equipment

	Item No	Options F 150
	351 1210 12	**Tool changer 24-position double arm gripper** Time - tool changing Tool to tool 1,8 Sek. (In exchange - Scope of delivery standard tool changer 16-position carousel type)
	351 1210 13	**Power transformer** for special tension
	351 1210 10	**Belt type of chip conveyor** (Retrofitting of scope of delivery chip conveyor screw-conveyor type)
	351 1210 11	**Chip carriage**
	351 1210 01	**Fourth axis Ø 120 mm** including assembly • incl. SIEMENS motor
	351 1210 06	**3-jaw chuck Ø 120 mm** for fourth axis
	351 1210 07	**Counterholder** for 3-jaw chuck Ø 120 mm for fourth axis
	351 1210 02	**Fifth axis Ø 100 mm** • incl. SIEMENS motors
	351121008	**Coolant channels** for spindles 20 bars integrated
	351 1210 09	**Coolant channels** for spindles 20 bars external version with external cooling tank
	351 12100 3	**Coolant channels** for spindles 70 bars external version with external cooling tank • Filtering accuracy 25 µm
	351 1210 05	**Air conditioning** (In exchange - Scope of delivery standard heat exchanger)
	351 1210 04	**Spindle 12'000 min^{-1}** (In exchange - Scope of delivery standard spindle 10'000 min^{-1})
	351 1230 12	**Tool Renishaw TS 27R**
	351 1230 17	**Machine reparation for Renishaw NC 4**
	351 1230 13	**Tool control system Renishaw NC4** • Contactless tool measurement and detection of tool break
	351 1230 18	**Machine reparation for Renishaw OMP60**
	351 1230 14	**Tool one measuring sensor Renishaw OMP60** • Optical signal transfer

Can only be ordered ex works.
For prices and further options please contact your dealer or
send us an e-mail: info@optimum-maschinen.de

The Optimum Premium CNC milling machine with SIEMENS control SINUMERIK 828 D distinguishes itself by performance, speed, accuracy and high durability

Convincing arguments: quality, efficiency and price

- Precision linear guideways in all axis
- 24-position magazine tool changer
- Cast iron body made of quality Meehanite cast iron
- Cross table precision ground with 5 T-slots
- Heavily ripped, torsion-free machine basis
- Precision ground pre-tensed high-performance ball screws in all axes Ø 40 x P12 X c3
- Main spindle SK40 up to 10'000 rev/min, with belt drive to minimise the vibrations as well as heat and noise development
- Spindle oil cooler
- Cross table and machine base made of high quality Meehanite cast iron, low-vibration
- Powerful servo drives directly on the ball screws in all 3 axes
- Chip conveyor incl. chip flushing system
- Cleaning gun
- Coolant pump with 4 bars
- Electronic handwheel
- Cleaning gun
- RJ45 plug connection, USB connection and power connection (230V)
- Telescopic guiding covers of all 3 axis
- Including 2-year Siemens warranty

Siemens Safety Integrated

Set up with opened door

safety integrated power by siemens

Including

Our options:

- **Safety Integrated**
- **Residual material detection and operation**
- **ShopMill Working step programming**
- **Managing network drive**
- **3-D simulation finished part**
- **Simultaneous recording**



PREMIUM

Type	F 310
Item No	351 1230
Electrical connection	
Connection ~50 Hz	400 V / 3 Ph
Power of the total connected value	25 kW
Driving motor	13 kW (S1)
Torque driving motor	70 Nm
Motor coolant pumps 4 pcs.	1.27 kW each
Spindle seat	
Spindle seat	ISO 40 DIN 69871
Milling head dimension max.	Ø 63 mm
End mill dimension max.	Ø 32 mm
Throat	615 mm
Milling accuracy	
Repeatability	0.005 mm
Positioning accuracy	± 0.005 mm
Tool changer	
Disc type	24 tools
Tool diameter max.	80 mm
Tool diameter max. (secondary places free)	125 mm
Tool length max.	300 mm
Tool weight max.	8 kg
Time - tool changing Tool to tool	1,8 seconds
Travels	
X axis	1'000 mm
Y axis	560 mm
Z axis	550 mm
Feed drive of axis	
Milling feed (X, Y, Z axis)	10'000 mm/min.
Torque motor	
X axis	16 Nm
Y axis	16 Nm
Z axis	11 Nm
Rapid feed	
X axis	30'000 mm/min
Y axis	30'000 mm/min
Z axis	24'000 mm/min
Speeds	
Speeds	10 - 10'000 min^{-1}
Cross table	
Distance spindle - table	100 - 650 mm
Spindle Z axis housing	615 mm
Table length x width	1'200 x 520 mm
T-slot size / number / distance	16 mm / 5 / 80 mm
Load (max.)	800 kg
Dimensions	
Length x Width x Height	4'080 x 2'600 x 3'130 mm
Tank capacity	210 litres
Total weight	6'020 kg

Standard equipment

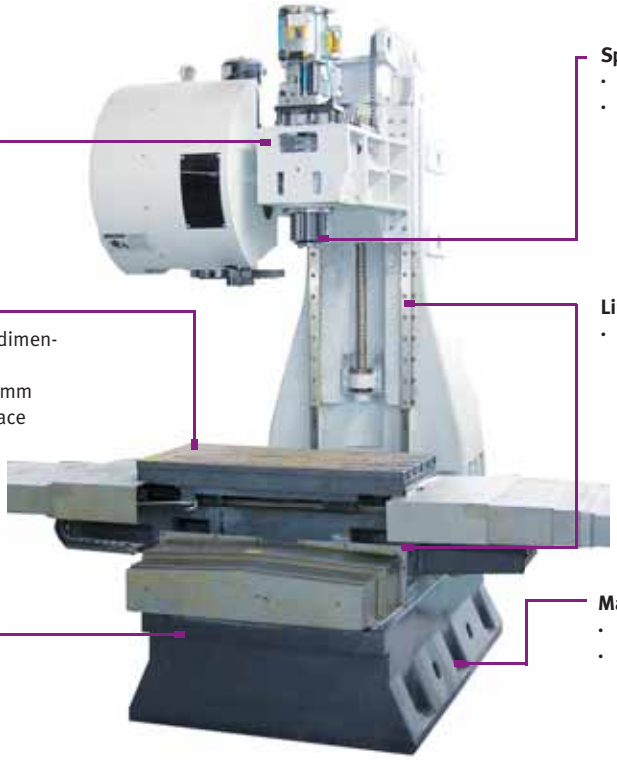
- Residual material detection and machining for contour pockets and stock removal
- Shop Mill
- Network connection
- 3-D simulation finished part
- Simultaneous recording
- Safety integrated
- Sinumerik Operate
- EMC
- Chip conveyor conveyor type
- Chip carriage
- Oil separator
- Heat exchanger
- Coolant pistol
- Chip flushing system
- Working lamp
- Tool changer 24-position double arm gripper
- Coolant pump 4 bars
- 6 pcs. Machine feet
- Spindle oil cooler

„Starter kit SK40/DIN 69871“ on page 79



„CNC-instruction - Information“ on page 80

Standard equipment



Cast iron body
- Quality Meehanite cast iron

Spindle
- ISO 40
- Spindle speed 10'000 min^{-1} as standard

Cross table
- Massive, accurate and largely dimensioned
- Clamping surface 1'200 x 528 mm
- Precisely machined at the surface
- Quality Meehanite cast iron

Linear guideways
- for high rapid feed speeds 30'000 min^{-1} in X- and Y- axis and 24'000 mm/min in the Z - axis

Base body
- Quality cast iron of ripped type

Machine feet
- 6 pieces
- Optimum alignment of the machine

Heat exchanger & Air Conditioner



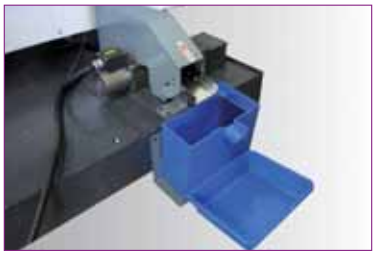
- Efficient integrated heat transfer from the switch cabinet
- Avoids that faults are generated from the overheated electrical and CNC components

Tool change system



- Automatic
- Magazine disc for 24 tools
- Max. tool length 300 mm

Oil separator



- Eliminating contaminants from the coolant

Handwheel



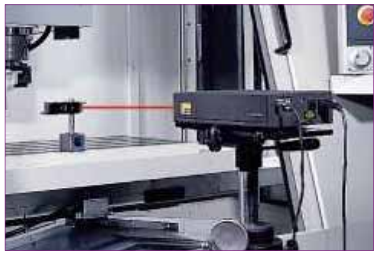
- Electronic system
- Reduces the set up time
- Emergency stop push button
- Acknowledgement button

Chip flushing system



- Powerful chip flush system to clean the working chamber and the workpiece

Laser measurement



- Guaranteed repeatability and positioning accuracy

Special equipment

	Item No	Options F 310
	351 1230 01	Fourth axis Ø 120 mm • incl. SIEMENS motor
	351 1230 06	**3-jaw chuck Ø 120 mm** for fourth axis
	351 1230 07	**Counterholder** for 3-jaw chuck Ø 120 mm for fourth axis
	351 1230 02	Fifth axis Ø 200 mm • incl. SIEMENS motor
	351 1230 08	**Coolant channels** for spindles 20 bars integrated
	351 1230 09	**Coolant channels** for spindles 20 bars external version with external cooling tank
	351 1230 03	**Coolant channels** for spindles 70 bars external version with external cooling tank • Filtering accuracy 25 µm
	351 1230 05	**Air conditioning** (Upgrade of the heat exchanger)
	351 1230 04	**Spindle 12'000 min^{-1}** (In exchange - Scope of delivery standard spindle 10'000 min^{-1})
	351 1230 10	**Tool changer 30-position double arm gripper** (In exchange - Scope of delivery standard tool changer 24-position double arm gripper)
	351 1230 11	**Chip conveyor screw-conveyor type** (In exchange - Scope of delivery standard chip conveyor belt type)
	351 1230 12	**Tool Renishaw TS 27R**
	351 1230 18	**Machine reparation for Renishaw NC 4**
	351 1230 13	**Tool control system Renishaw NC4** • Contactless tool measurement and detection of tool break
	351 1230 18	**Machine reparation for Renishaw OMP60**
	351 1230 14	**Tool one measuring sensor Renishaw OMP60** • Optical signal transfer

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send us an e-mail: info@optimum-maschinen.de

The Optimum Premium CNC milling machine with SIEMENS control SINUMERIK 828 D distinguishes itself by performance, speed, accuracy and high durability

Convincing arguments: quality, efficiency and price

- Precision linear guideways in all axis
- 24-position magazine tool changer
- Cast iron body made of quality MEEHANITE cast iron
- Cross table precision ground with 5 T-slots
- Heavily ripped, torsion-free machine basis
- Precision ground pre-tensed high-performance ball screws in all axes Ø 40 x P12 X c3
- Main spindle SK40 up to 10000 rev/min, with belt drive to minimise the vibrations as well as heat and noise development
- Spindle oil cooler
- Table and machine basis made of high quality Meehanite cast iron, low-vibration
- Powerful servo drives directly on the ball screws in all 3 axes
- Chip conveyor incl. chip flushing system
- Cleaning gun
- Coolant pump with 4 bars
- Electronic handwheel
- Cleaning gun
- RJ45 plug connection, USB connection and power connection (230V)
- Including 2-year Siemens warranty

PREMIUM

Siemens Safety Integrated

Set up with opened door

safety integrated power by siemens

Including

Our options:

- **Safety Integrated**
- **Residual material detection and operation**
- **ShopMill Working step programming**
- **Managing network drive**
- **3-D simulation finished part**
- **Simultaneous recording**



Fig.: F 410

Type	F 410
Item No	351 1240
Electrical connection	
Connection ~50 Hz	400 V / 3 Ph
Power of the total connected value	35 kW
Driving motor	13 kW (S1)
Torque driving motor	70 Nm
Motor coolant pumps 2 pcs.	1.27 kW each
Motor coolant pump 1 pc.	750 W
Spindle seat	
Spindle seat	ISO 40 DIN 69871
Milling head dimension max.	Ø 63 mm
End mill dimension max.	Ø 32 mm
Milling accuracy	
Repeatability	0.005 mm
Positioning accuracy	± 0.005 mm
Tool changer	
Disc type	24 tools
Tool size max.	130 mm
Tool diameter max.	80 mm
Tool diameter max. (secondary places free)	125 mm
Tool length max.	300 mm
Tool weight max.	8 kg
Time - tool changing; Tool to tool	1,8 seconds
Travels	
X axis	1'300 mm
Y axis	680 mm
Z axis	730 mm
Feed drive of axis	
Milling feed (X, Y, Z axis)	10'000 mm/min.
Torque motor	
X axis	18 Nm
Y axis	18 Nm
Z axis	16 Nm
Rapid feed	
X axis	30'000 mm/min
Y axis	30'000 mm/min
Z axis	24'000 mm/min
Speeds	
Speeds	10 - 10'000 min^{-1}
Cross table	
Distance spindle - table	100 - 830 mm
Table length x width	1'700 x 600 mm
T-slot size / number / distance	18 mm / 5 / 100 mm
Load (max.)	1'000 kg
Dimensions	
Length x Width x Height	4'200 x 2'720 x 3'130 mm
Tank capacity	210 litres
Total weight	8'480 kg

„Starter kit SK40/DIN 69871“ on page 79

Standard equipment

- Residual material detection and machining for contour pockets and stock removal
- Shop Mill
- Network connection
- 3-D simulation finished part
- Simultaneous recording
- Safety integrated
- EMC
- Sinumerik Operate
- Chip conveyor belt type
- Chip carriage
- Oil separator
- Heat exchanger
- Coolant pistol
- Chip flushing system
- Working lamp
- Tool changer 24-position double arm gripper
- Coolant pump 4 bars
- 6 pcs. Machine feet
- Spindle oil cooler

Spindle oil cooler

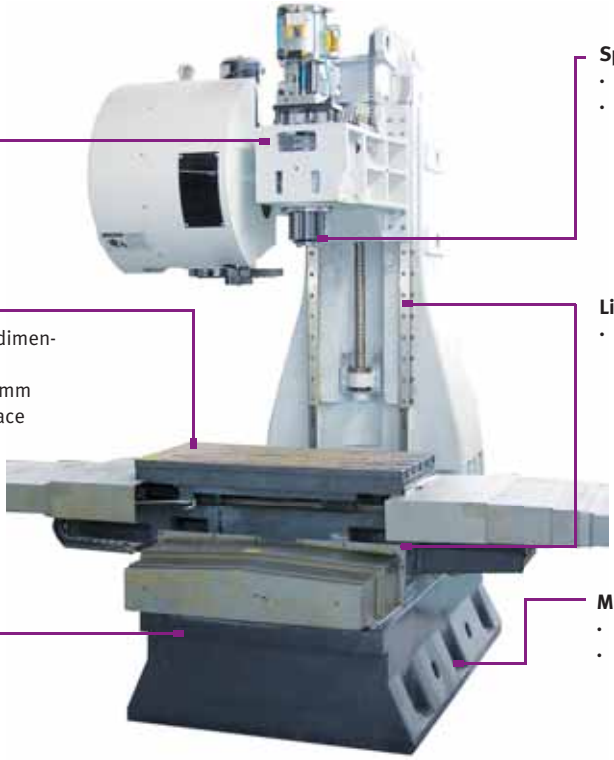


The machine is already equipped as standard with max. spindle speed 10'000 min^{-1} and spindle oil cooler



„CNC-instruction - Information“ on page 80

Standard equipment



Cast iron body
- Quality Meehanite cast iron

Spindle
- ISO 40
- Spindle speed 10'000 min^{-1} as standard

Cross table
- Massive, accurate and largely dimensioned
- Clamping surface 1'200 x 528 mm
- Precisely machined at the surface
- Quality Meehanite cast iron

Linear guideways
- for high rapid feed speeds 30'000 min^{-1} in X- and Y- axis and 24'000 mm/min in the Z - axis

Base body
- Quality cast iron of ripped type

Machine feet
- 6 pieces
- Optimum alignment of the machine

Heat exchanger & Air Conditioner



- Efficient integrated heat transfer from the switch cabinet
- Avoids that faults are generated from the overheated electrical and CNC components

Tool change system



- Automatic by means of magazine disc for 24 tools
- Max. tool length 300 mm

Chip carriage



- tiltable
- rollable

Handwheel



- Electronic system
- Reduces the set up time
- Emergency-stop push button
- Acknowledgement button

Chip flushing system



- Efficient disposal of chips
- keeps the work area free from chips

Laser measurement



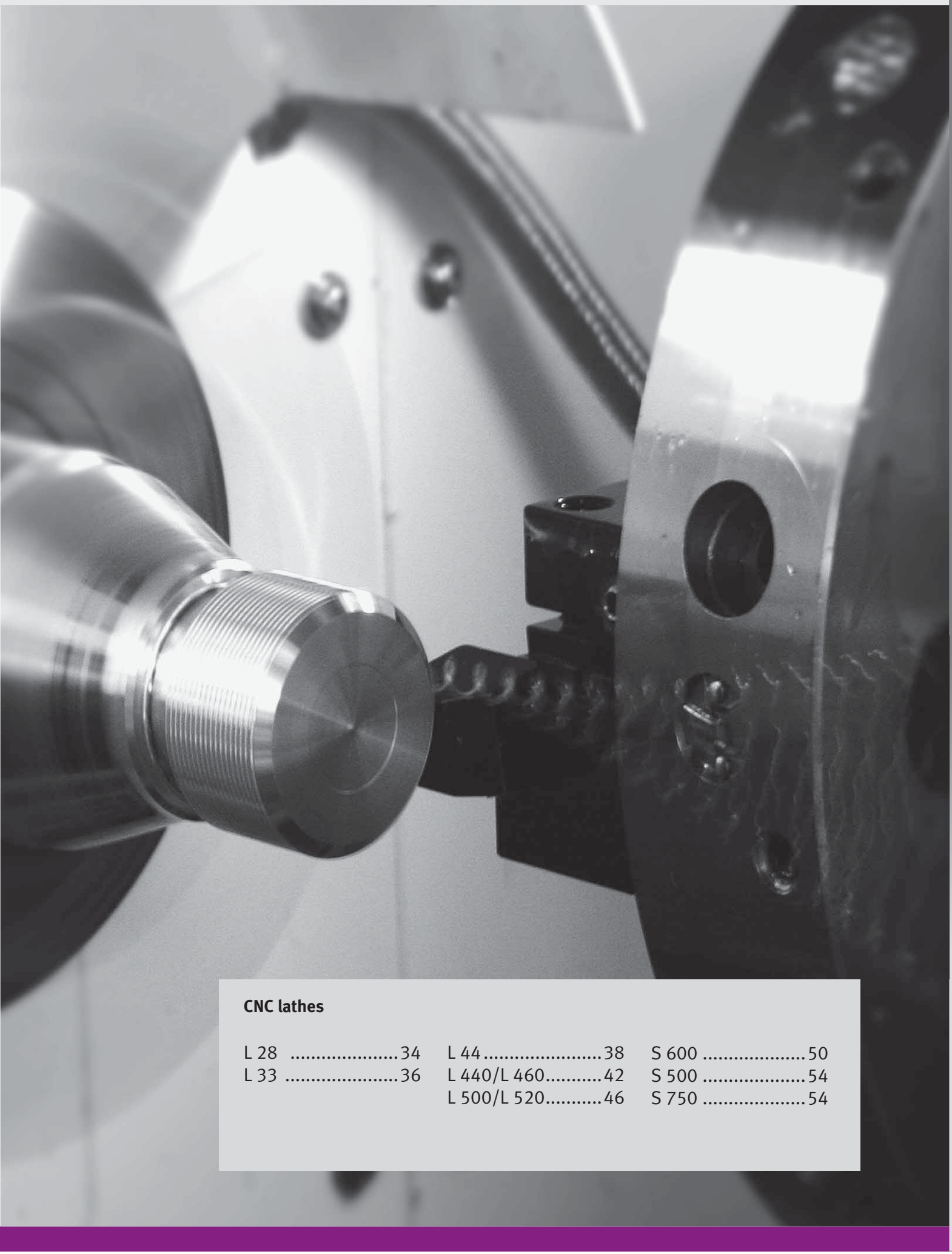
- Guaranteed repeatability and positioning accuracy

Special equipment

	Item No	Options F 410
	351 1240 01	**Fourth axis Ø 120 mm** • incl. SIEMENS motor
	351 1240 06	**3-jaw chuck Ø 120 mm** for fourth axis
	351 1240 07	**Counterholder** for 3-jaw chuck Ø 120 mm for fourth axis
	351 1240 02	**Fifth axis Ø 200 mm** including • incl. SIEMENS motor
	351 1240 08	**Coolant channels** for spindles 20 bars integrated
	351 1240 09	**Coolant channels** for spindles 20 bars external version with external cooling tank
	351 1240 03	**Coolant channels** for spindles 70 bars external version with external cooling tank • Filtering accuracy 25 µm
	351 1240 05	**Air conditioning** (Upgrade of the heat exchanger)
	351 1240 04	**Spindle 12'000 min^{-1}** (In exchange - Scope of delivery standard spindle 10'000 min^{-1})
	351 1240 10	**Tool changer 30-position double arm gripper** (In exchange - Scope of delivery standard tool changer 24-position double arm gripper)
	351 1240 11	**Chip conveyor screw-conveyor type** (In exchange - Scope of delivery standard chip conveyor belt type)
	351 1240 12	**Tool Renishaw TS 27R**
	351 1240 18	**Machine reparation for Renishaw NC 4**
	351 1240 13	**Tool control system Renishaw NC4** • Contactless tool measurement and detection of tool break
	351 1240 17	**Machine reparation for Renishaw OMP40**
	351 1240 14	**Tool one measuring sensor Renishaw OMP60** • Optical signal transfer

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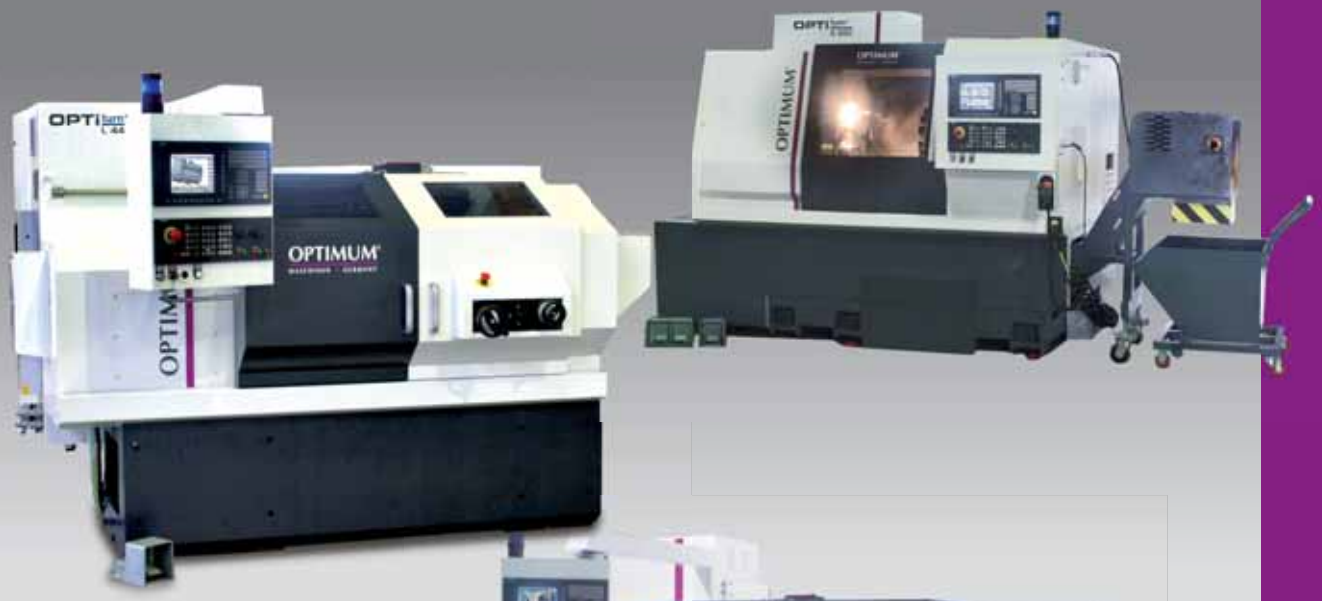
„CNC-instruction - Information“ on page 80



CNC lathes

L 28 34
L 33 36
L 44 38
L 440/L 460 42
L 500/L 520 46
S 600 50
S 500 54
S 750 54

CNC lathes





CNC controlled flat bed lathe equipped with SIEMENS control, perfect for training purposes

Convincing arguments: quality, efficiency and price

- Braced machine bed made of grey cast iron
- Bed guideways inductively hardened (HRC 42-52) and precision ground
- Elaborate spindle bearing
- Guaranteed concentricity of the spindle nose less than 0.009 mm
- Concentricity on the lathe chuck less than 0.04 mm
- Maintenance-free motor
- Brushless DC drive, infinitely variable
- Emergency-stop push button
- Central lubrication
- Incremental position encoder for spindle positioning (thread cutting)
- Right-/left handed rotation
- Ball screw made by HIWIN
- Reference switch
- Easy to maintain protective housing
- Rear access flap for maintenance purposes
- Safety switch at the front sliding door

SIEMENS SINUMERIK 802S base line

SIEMENS control
- in a slewable housing
- Two electronic handwheels allow manual working

- Automatic quadruple tool changer for small series
- Servo motor equipped with worm gear
- Tool change controlled by CNC program



Standard equipment



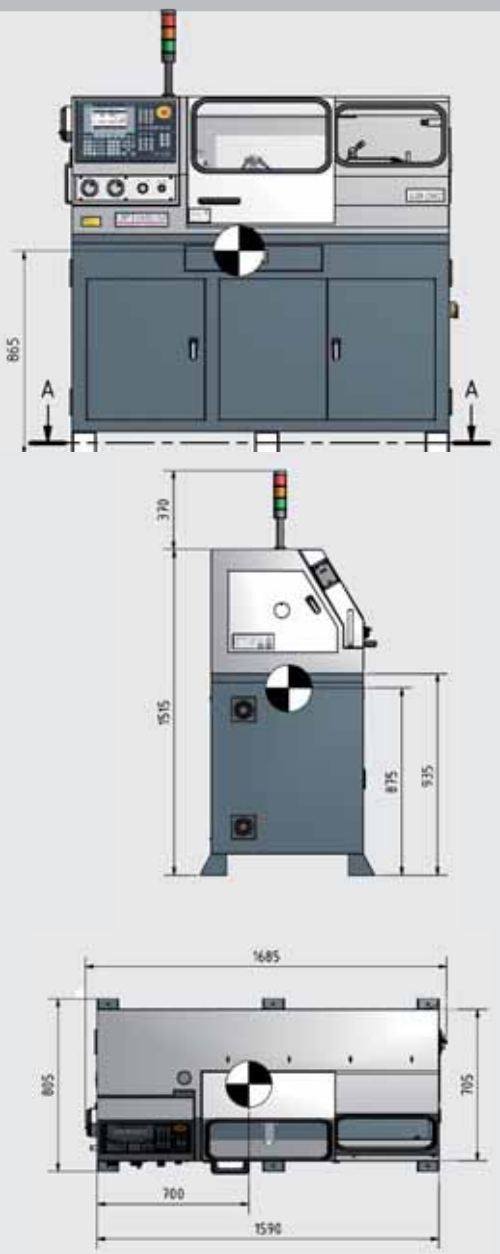
- Controller IV
- Control of the axis drives



- Base with coolant system
- Tank capacity 12 litres
- Extendable chip drawer
- Telescopic guide rail
- Coolant drain

Type	L 28
Item No	350 4110
Electrical connection	
Total connection	2.7 kW 230 V ~50 Hz
Driving motor	1.5 kW
Power of the coolant pump	40 W
Tank capacity	12 litres
Machine data	
Centre height	140 mm
Centre width	700 mm
Turning Ø over machine bed	266 mm
Bed width	180 mm
Speeds	
Spindle speed	30 - 4'000 min^{-1}
Speed stages	4 steps, infinitely variable
Spindle	
Spindle taper	MT 4
Spindle bore	Ø 26 mm
Quadruple tool holder seat height	(max.) 12 mm
Tool changer	
Number of tool positions	4
Seat height max.	12 mm
Travels	
X axis	180 mm
Z axis	465 mm
Feed speed	
X axis	3'000 mm/min.
Z axis	2'500 mm/min.
Tailstock	
Tailstock seat	MT 2
Tailstock - sleeve travel	80 mm
Dimensions	
Length x Width x Height (open)	1'885 x 805 x 1'685 mm
Total weight	420 kg

Dimensions



Stage	1	2	4	3
Speed [min^{-1}]	30 - 300	140 - 1400	300 - 3000	400 - 4000
Torque [Nm]	98	21	9.8	7

Standard equipment

- 3-jaw chuck Ø 125 mm
- Steady centre point MT2/MT4
- Coolant equipment
- Machine lamp
- Operating tool



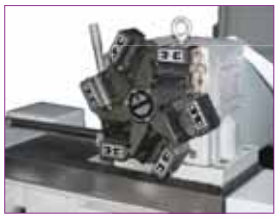
- We imperatively recommend the briefing performed by our specialists.

„CNC-instruction - Information“ on page 80

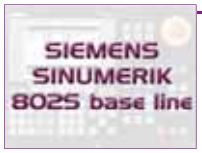
CNC controlled flat bed lathe equipped with SIEMENS control, perfect for small series and training purposes

Convincing arguments: quality, efficiency and price

- Precise machining
- Braced machine bed made of grey cast iron
- Bed guideways inductively hardened (HRC 42-52) and precision ground
- Elaborate spindle bearing
- Emergency-stop push button
- Central lubrication
- Incremental position encoder for spindle positioning (thread cutting)
- Right-/left handed rotation
- Grinded ball screw spindles
- Reference switch
- Two-stage gear ratio for a powerful transmission of the motor power
- Easy to maintain protective housing
- Rear access flap for maintenance purposes
- Safety switch at the front sliding door
- SIEMENS Micromaster as frequency converter
- Control via Controller IV



- Automatic 6-fold tool changer for serial production
- High repeat accuracy



SIEMENS control
- in a slewable housing
- Two electronic handwheels allow manual working



Standard equipment



Motor spindle with encoder
- Readings recorder for rotary movements
- for highest pitch accuracy

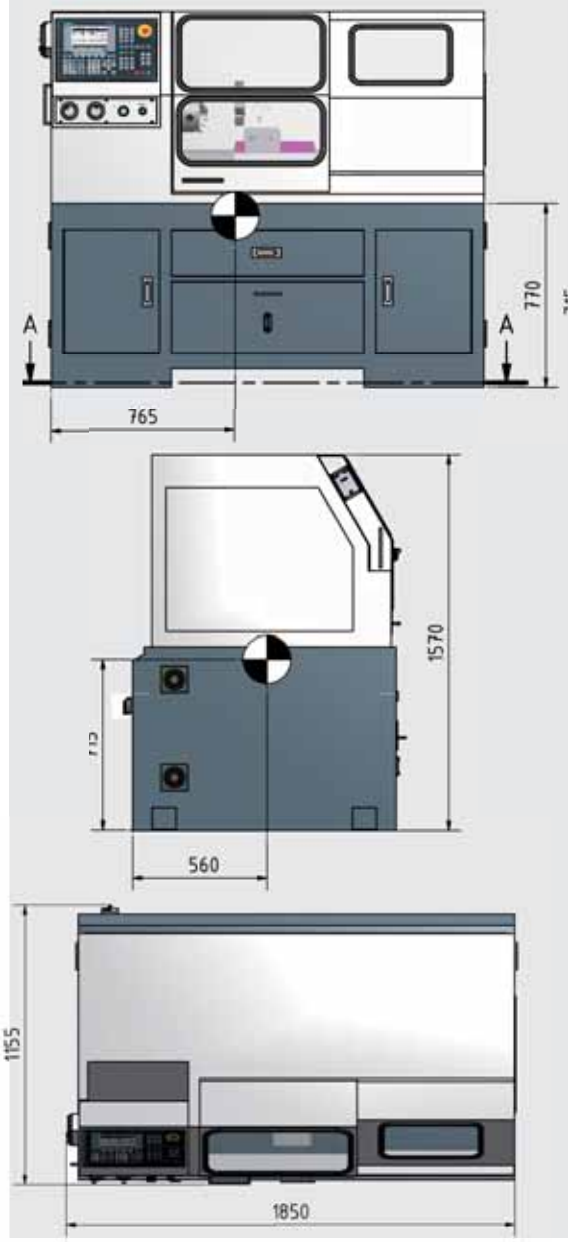


Cable duct
- centrally via energy chain

Reference switch
- of mechanic type

Type	L 33
Item No	350 4115
Electrical connection	
Total connection	3.5 kW 400 V ~50 Hz
Drive motor	3 kW
Power of the coolant pump	40 W
Tank capacity	50 litres
Machine data	
Centre height	165 mm
Centre width	750 mm
Turning Ø over cross slide	140 mm
Turning Ø over machine bed	330 mm
Bed width	187 mm
Speeds	
Spindle speed "L"	80 - 1'200 min^{-1}
Spindle speed "H"	500 - 3'000 min^{-1}
Torque spindle motor "L"	36 Nm
Torque spindle motor "H"	16 Nm
Spindle	
Spindle seat	Camlock D 1-4"
Spindle bore	Ø 38 mm
Quadruple tool holder seat height	(max.) 16 mm
Tool changer	
Number of tool positions	6
Seat height max.	16 mm
Travel	
X axis	165 mm
Z axis	600 mm
Feed speed	
The indicated speeds were determined in the 1/16 step process	
X axis	2'000 mm/min.
Z axis	2'000 mm/min.
Pitch of ball screw	
X axis	4 mm
Z axis	6 mm
Torque motor	
X axis	8.4 Nm
Z axis	10 Nm
Tailstock	
Tailstock seat	MT 3
Tailstock - sleeve travel	120 mm
Dimensions	
Length x Width x Height (open)	1'860 x 1'160 x 1'580 mm
Total weight	1'000 kg

Dimensions



Standard equipment

- 3-jaw chuck Ø 160 mm centric clamping
- Coolant equipment
- Machine lamp
- Operating tool



„CNC-instruction - Information“ on page 80

Optimum CNC lathe with SIEMENS control 828 Basic T and the advantages: Speed, power, accuracy and durability

Convincing arguments: quality, efficiency and price

- Spindle and servo motor made by SIEMENS
- Complete cladding with safety equipment
- Coolant equipment
- Automatic central lubrication
- High spindle speed
- Slewable operating unit
- Electronic handwheels for X and Z axis
- RJ45 plug connection, USB connection and power connection (230V)
- Two-year SIEMENS guaranty included



Fig. including optional hydraulic tailstock spindle sleeve

Siemens Safety integrated

Set up with opened door

safety integrated power by siemens

PREMIUM

Standard equipment

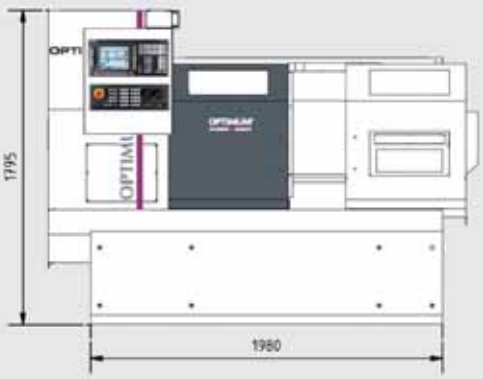
Residual material detection and machining for contour pockets and stock removal
Shop Turn
Network drive management
3-D simulation finished part
Simultaneous recording
Safety integrated
EMC
Joystick
Electronic handwheel
Hydraulic turret Type VDI 30 (8 tools)
3-jaw chuck hydraulic Ø 150 mm
Hard and non-rigid block jaws
Heat exchanger
Tailstock end cover
6 pcs. Machine feet
Operating tool

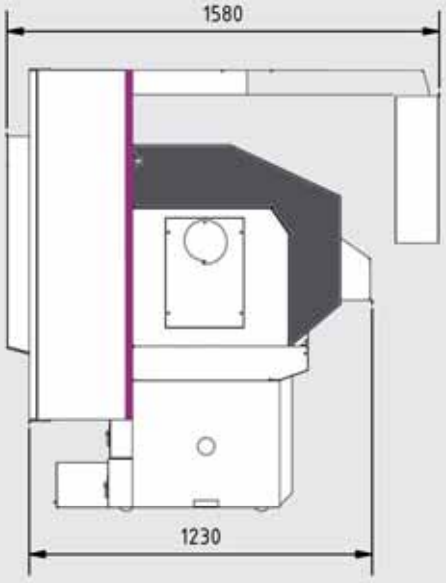
Type	L 44
Item No	351 4330
Electrical connection	
Total connection	18.5 kW 400 V ~50 Hz
Driving motor	8 kW (S1 mode 7kW)
Torque drive motor	31 Nm
Power of the coolant pump	270 W
Tank capacity	90 litres
Hydraulic system	
Motor power	750 W
Tank capacity	50 litres
Machine data	
Centre height	223 mm
Centre width	850 mm
Turning Ø over cross slide	240 mm
Turning Ø over machine bed	446 mm
Turning Ø in the bed bridge	520 mm
Speeds	
Spindle speed, 2 steps	80 - 3'200 min^{-1}
Spindle	
Spindle seat	DIN ISO 702-1 Nr. 5
Spindle bore	Ø 52 mm
Hydraulic lathe chuck	Ø 150 mm
Chuck through hole	Ø 40 mm
Bed width	300 mm
Turret	
Hydraulic type	VDI 30
Number of tool positions	8
Seat height, seat width max.	20 x 20 mm
Seat diameter max.	Ø 25 mm
Accuracy	
Repeatability	± 0.005 mm
Positioning accuracy	± 0.005 mm
Travels	
X axis	250 mm
Z axis	760 mm
Feed speed	
X axis	15'000 mm/min
Z axis	15'000 mm/min
Torque motor	
X axis	6 Nm
Z axis	8.5 Nm
Ball screw	
X axis	25 mm × P5 mm × C3
Z axis	40 mm × P5 mm × C3
Tailstock	
Tailstock seat	MT 4
Tailstock - sleeve diameter	52 mm
Tailstock - sleeve travel	165 mm
Dimensions	
Length x Width x Height (open)	2'230 x 1'580 x 1'795 mm
Total weight	2'100 kg

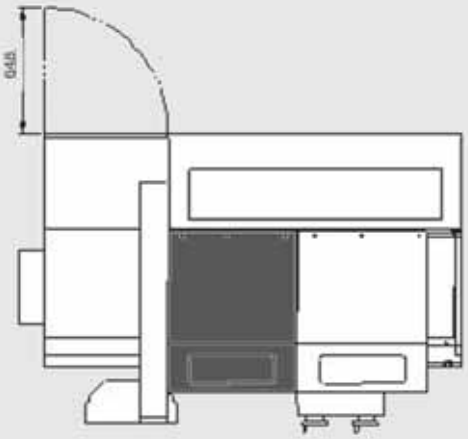


„CNC-instruction - Information“ on page 80

Dimensions







„Starter kit VDI 30“ on page 74

Standard equipment

Main spindle
- Massive type

Machine bed
- Deeply ribbed
- High stiffness and durability



Guiding
- Stable flat bed guiding

Ball screw
- ground

Cast iron body
- Hardened (HRC48-52) and heat treated

Machine feet
- **6 pcs**
- for optimum alignment of the machine

Heat exchanger



- for efficient integrated heat transfer from the switch cabinet
- Avoids that faults are generated from the overheated electrical and CNC components

Tool change system



- 8 tool positions
- Hydraulic VDI30 turret
- Seat height max. 20 mm

Tailstock



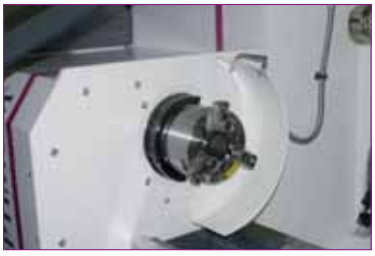
- generously dimensioned
- Carriage with ball screw with prestressed nut

Hydraulic unit



- Tank capacity 50 litres

Three-jaw chuck



- Hydraulic three-jaw chuck Ø 150 mm
- Simple clamping of workpieces

Central lubrication system



- Automatically
- Fold-out for easy filling
- The floating switch identifies the volume of the oil and releases an automatic alarm as soon as it falls below the set value

Special equipment

	Item No	Options L 44
	351 4330 06	**Non-rigid block jaws** for scope of delivery hydraulic 3-jaw lathe chuck Ø 150 mm
	351 4330 05	**Hard block jaws** for scope of delivery hydraulic 3-jaw lathe chuck Ø 150 mm
	351 4330 15	**3-jaw lathe chuck Ø 200 mm** (In exchange - Scope of delivery 3-jaw chuck hydraulic Ø 150 mm)
	351 4330 17	**Non-rigid block jaws** for 3-jaw lathe chuck Ø 200 mm
	351 4330 16	**4-jaw lathe chuck Ø 250 mm** (In exchange - Scope of delivery hydraulic 3-jaw chuck Ø 150 mm)
	351 4330 18	**Hydraulic 3-jaw lathe chuck Ø 200 mm** (Surcharge in exchange - Scope of delivery hydraulic 3-jaw chuck Ø 150 mm)
	351 4330 20	**Hydraulic 4-jaw lathe chuck Ø 200 mm** (In exchange - Scope of delivery 3-jaw chuck hydraulic Ø 150 mm)
	351 4330 02	**Steady rest** 10 mm - 130 mm
	351 4330 03	**Follow rest** 10 mm - 100 mm
	351 4330 10	**Working lamp**
	351 4330 07	**LS-120 Hydraulic turret** Standard type - incl. 8 tools
	351 4330 08	**Tool holder kit** for LS-120 hydraulic turret
	351 4330 11	**Quick action tool holder Multifix4** (In exchange - Scope of delivery hydraulic turret LS-120)
	351 4330 09	**Portable handwheel** (In exchange - Standard joystick)
	351 4330 01	**Hydraulic tailstock spindle sleeve**
	351 4330 12	**Bar feeder PRO 1.25 mtr.**
	351 4330 19	**Bar feeder PRO 3.20 mtr.** for diameter 5 mm - 51 mm
	351 4330 13	**Power transformer** for special tension

Can only be ordered ex works.
For prices and further options
please contact your dealer or
send us an e-mail:
info@optimum-maschinen.de

OPTIMUM Premium high-capacity CNC cycle lathes incl. SIEMENS control Sinumerik 828D Basic T

Convincing arguments: quality, efficiency and price

- High precision machine with latest SIEMENS control and SIEMENS servo drives
- New compact structure of the headstock
- High-speed spindle (4500 rev/min) with high-precise, largely dimensioned taper roller bearings
- Change-over between the two gears is performed pneumatically via the air cylinder
- Headstock structure guarantees minimum noise development
- Wide machine bed with double rectangle slideways also for roughing work
- Hardened and ground slide ways
- Largely dimensioned tailstock, travelling on two guideways, easy-to-position via carriage and is equipped with a quick clamping
- 2 front separately travelling sliding doors incl. integrated windows
- Micro switches prevent starting up the machine if the protecting device is not completely closed and thus avoids the opening of the door
- RJ45 plug connection, USB connection and power connection (230V)
- Travelling operating panel
- Two electronic handwheels for manual operation of the X and Z axis
- Automatic lubrication of the headstock
- Automatic lubrication of the saddle slide and the cross slide rest
- Halogen working lamp
- Heat exchanger for switch cabinet
- Two-year SIEMENS guaranty included

Set up with opened door

safety integrated power by siemens



PREMIUM

Standard equipment	
Residual material detection and machining for contour pockets and stock removal	Hydraulic turret LS160 VDI40
Shop Turn	3-jaw chuck hydraulic Ø 200 mm
Network drive management	Hard and non-rigid block jaws
3-D simulation finished part	Heat exchanger
Simultaneous recording	Tailstock end cover
Safety integrated	6 pcs. Machine feet
EMC	Coolant equipment
Electronic handwheel	

Type	L 440	L 460
Item No	351 4410	351 4420
Electrical connection		
Total connection	20 kW 400 V ~50 Hz	20 kW 400 V ~50 Hz
Driving motor	12 kW	12 kW
Torque drive motor	70 Nm	70 Nm
Power of the coolant pump	560 W	560 W
Tank capacity (without chip conveyor)	170 litres	240 litres
Hydraulic system		
Motor power	750 W	750 W
Tank capacity	50 litres	50 litres
Machine data		
Centre height	235 mm	235 mm
Cutting length max.	1'000 mm	1'500 mm
Turning Ø over cross slide	240 mm	240 mm
Turning Ø over machine bed	475 mm	475 mm
Turning Ø in the bed bridge	710 mm	710 mm
Speeds		
Spindle speed (chuck-Ø 160 mm)	100 – 4'500 min^{-1}	100 – 4'500 min^{-1}
Spindle		
Spindle seat	DIN ISO 702-1 Nr. 6	DIN ISO 702-1 Nr. 6
Spindle bore	Ø 65 mm	Ø 65 mm
Hydraulic lathe chuck	Ø 200 mm	Ø 200 mm
Chuck through hole	Ø 52 mm	Ø 52 mm
Turret		
Hydraulic type	VDI 40	VDI 40
Number of tool positions	8	8
Seat height, seat width max.	25 x 25 mm	25 x 25 mm
Seat diameter max.	Ø 32 mm	Ø 32 mm
Accuracy		
Repeatability	± 0.005 mm	± 0.005 mm
Positioning accuracy	± 0.005 mm	± 0.005 mm
Travels		
X axis	260 mm	260 mm
Z axis	1'150 mm	1'680 mm
Feed speed		
X axis	15'000 mm/min	15'000 mm/min
Z axis	15'000 mm/min	15'000 mm/min
Torque motor		
X axis	6 Nm	6 Nm
Z axis	16 Nm	16 Nm
Tailstock		
Tailstock seat	MT 4	MT 4
Tailstock - sleeve diameter/travel	65 mm / 150 mm	65 mm / 150 mm
Dimensions		
Length x Width x Height	3'030 x 1'952 x 2'025 mm	3'530 x 1'952 x 2'025 mm
Total weight	3'000 kg	3'450 kg

Information

Lifting device

Item No. 351 4301



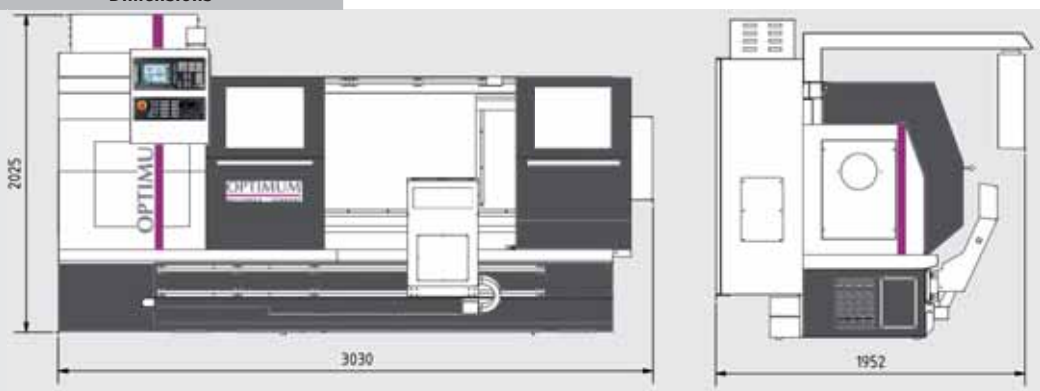
- A lifting device is required to unload the machine. The company Optimum may make it available by way of loan on bail in Germany and Austria. It may optionally be purchased.



„CNC-instruction - Information“ on page 80

„Starter kit VDI 40“ on page 75

Dimensions



Standard equipment

Machine bed



- Machine bed is designed particularly wide
- Ball screw drive for higher repeatability

Headstock



- Precision bearings
- Stable structure
- Smooth running also at high spindle speeds

Heat exchanger



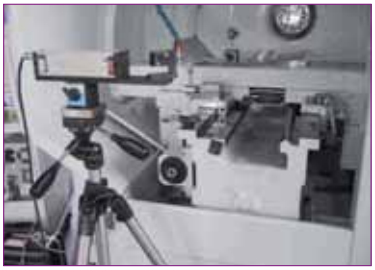
- Efficient integrated heat transfer from the switch cabinet
- Avoids that faults are generated from the overheated electrical and CNC components

Rear side of the machine



- Large opening at the rear side allows easy access for maintenance purposes

Laser measurement



- Guaranteed repeatability and positioning accuracy

Turret



- 8 tool positions
- Hydraulic - VDI40
- Seat height max. 25 mm

Tailstock



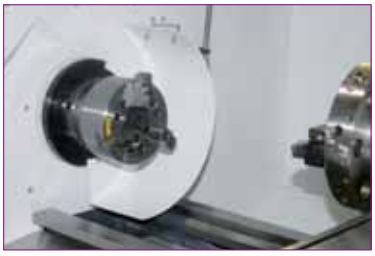
- Generously dimensioned
- Carriage with ball screw with prestressed nut

Hydraulic unit



- Tank capacity 50 litres

Three-jaw chuck



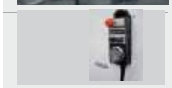
- Hydraulic three-jaw chuck Ø 200 mm
- Simple clamping of workpieces

Coolant equipment



- High power coolant pump 375 W
- Tank capacity 100 litres

Special equipment

	Item No L 460	Item No L 440	Options
	351 4420 08	351 4410 08	**3-jaw lathe chuck Ø 200 mm** manual
	351 4420 10	351 4410 10	**Base jaws and non-rigid block jaws** for 3-jaw lathe chuck Ø 200 mm
	351 4420 28	351 4410 28	**Hydraulic 3-jaw lathe chuck Ø 250 mm** (In exchange - Scope of delivery hydraulic 3-jaw lathe chuck Ø 200 mm)
	351 4420 11	351 4410 11	**4-jaw lathe chuck Ø 250 mm**
	351 4420 31	351 4410 31	**Hydraulic 4-jaw lathe chuck Ø 250 mm** (In exchange - Scope of delivery hydraulic 3-jaw lathe chuck Ø 200 mm)
	351 4420 02	351 4410 02	**Joystick** to travel the X- and Z axis (not combinable with electronic handwheel)
	351 4420 20	351 4410 20	Detachable **electronic handwheel** (not combinable with joystick)
	351 4420 07	351 4410 07	**Oil separator**
	351 4420 05	351 4410 05	**Chip conveyor** 1.0 meter (L 440) 1.50 meter (L 460)
	351 4410 06	351 4410 06	Chip carriage
	351 4420 12	351 4410 12	**Turret Baruffaldi TB 160 VDI40** 8 tools - driven tools (In exchange - Scope of delivery standard turret LS-160)
	351 4420 17	351 4410 17	**C axis brake system** (only with turret Baruffaldi TB 160)
	351 4420 16	351 4410 16	**Tool holder kit** for LS-160 turret
	351 4420 13	351 4410 13	**Coolant pump** 5 bars
	351 4420 01	351 4410 01	**Coolant channels** for tools 20 bars
	351 4420 14	351 4410 14	**Air conditioning** for switch cabinet (In exchange - Scope of delivery standard heat exchanger)
	351 4420 03	351441003	**Steady rest** 20 mm - 200 mm
	351 4420 04	351 4410 04	**Follow rest** 20 mm - 100 mm
	351 4420 19	351 4410 19	**Preparation for hydraulic tailstock spindle**
	351 4420 21	351 4410 21	**Hydraulic tailstock spindle sleeve**
	351 4420 22	351 4410 22	**Device to travel the tailstock simultaneously**
	351 4420 23	351 4410 23	**Pneumatic lifting device for tailstock**
	351 4420 33	351 4410 33	**Bar feeder 1,25 mtr.** V65-E-Pro
	351 4420 34	351 4410 34	**Bar feeder 1,5 mtr.** V65-LE-Pro
	351 4320 35	351 4310 35	**Bar feeder 1,25 mtr.** DH 65 Fedek
	351 4320 36	351 4310 36	**Bar feeder 1,5 mtr.** DH 65L Fedek

Can only be ordered ex works. For prices and other options please contact your dealer or send us an e-mail to: info@optimum-maschinen.de

Optimum Premium high-capacity CNC cycle lathes incl. SIEMENS control Sinumerik 828 Basic T

Convincing arguments: quality, efficiency and price

- High precision machine with latest SIEMENS control and SIEMENS servo drives
- New compact structure of the headstock
- High-speed spindle (3'500 rev/min) with high-precise, largely dimensioned taper roller bearings
- Change-over between the two gears is performed pneumatically via the air cylinder
- Headstock structure guarantees minimum noise development
- At high speeds a thermal expansion is almost avoided contrary to the pure gear drive.
- Stable ball screw with a diameter of 45 mm provides for longitudinal movements
- The hardened and precision-ground ball bearings of the X and Z axis mounted in high precision bearings allow most accurate traverse movements
- Machine bed with double rectangular guiding designed for high rapid traverse
- Hardened and ground slide ways
- Largely dimensioned tailstock, travelling on two guideways, easy-to-position via carriage equipped with quick clamping
- 2 front separately travelling sliding doors incl. integrated window
- Micro switches prevent starting up the machine if the protecting device is not completely closed and thus avoids opening
- RJ45 plug connection, USB connection and power connection (230V)
- Travelling operating panel
- 2 electronic handwheels for manual operation of the X and Z axis
- Automatic lubrication of the headstock
- Automatic lubrication of the saddle slide and the cross slide rest
- Halogen working lamp
- Two-year SIEMENS guaranty included

SIEMENS SINUMERIK 828 D Basic T

Siemens Safety Integrated

Set up with opened door

safety integrated power by siemens

PREMIUM



Standard equipment	
Residual material detection and machining for contour pockets and stock removal	Hydraulic turret LS160 VDI40
Shop Turn	3-jaw chuck hydraulic Ø 250 mm
Network drive management	Hard and non-rigid block jaws
3-D simulation finished part	Heat exchanger
Simultaneous recording	Tailstock end cover
Safety integrated	6 pcs. Machine feet
EMC	Siemens Tool Box CD
Electronic handwheel	Chip carriage
Tailstock locking	Chip conveyor

Type	L 500	L 520
Item No	351 4430	351 4440
Electrical connection		
Total connection	20 kW 400 V ~50 Hz	30 kW 400 V ~50 Hz
Driving motor	12 kW (S1)	17.5 kW (S1)
Torque drive motor	115 Nm	115 Nm
Power of the coolant pump	450 W	450 W
Tank capacity	208 litres	260 litres
Hydraulic system		
Motor power	750 W	750 W
Tank capacity	50 litres	50 litres
Machine data		
Centre height	235 mm	235 mm
Distance between centres max.	2'000 mm	3'000 mm
Turning Ø over cross slide	310 mm	310 mm
Turning Ø over machine bed	550 mm	550 mm
Turning Ø in the bed bridge	790 mm	790 mm
Max. workpiece weight (incl. tailstock)	2'000 kg	2'000 kg
Bed width	405 mm	405 mm
Speeds		
Spindle speeds	80 – 3'500 min^{-1}	80 – 3'500 min^{-1}
Spindle		
Spindle seat	DIN ISO 702-1 Nr. 8	DIN ISO 702-1 Nr. 8
Spindle bore	Ø 82 mm	Ø 82 mm
Hydraulic lathe chuck	Ø 250 mm	Ø 250 mm
Chuck through hole	Ø 69 mm	Ø 69 mm
Turret		
Hydraulic type	VDI 40 (5480)	VDI 40 (5480)
Number of tool positions	12	12
Seat height, seat width max.	25 x 25 mm	25 x 25 mm
Seat diameter max.	Ø 32 mm	Ø 32 mm
Accuracy		
Repeatability	± 0.005 mm	± 0.005 mm
Positioning accuracy	± 0.005 mm	± 0.005 mm
Travels		
X axis	345 mm	345 mm
Z axis	2'150 mm	3'150 mm
Feed speed		
X axis	15'000 mm/min	15'000 mm/min
Z axis	15'000 mm/min	15'000 mm/min
Torque motor		
X axis	11 Nm	11 Nm
Z axis	20 Nm	20 Nm
Tailstock		
Tailstock seat	MT 5	MT 5
Tailstock - sleeve diameter	80 mm	80 mm
Tailstock - sleeve travel	150 mm	150 mm
Dimensions		
Length x Width x Height	4'190 x 2'245 x 2'050 mm	5'190 x 2'245 x 2'050 mm
Total weight	5'170 kg	6'200 kg

Scope of delivery

- Chip carriage
- Chip conveyor
- Non-rigid and hard block jaws

Information

Lifting device

Item No. 351 4302



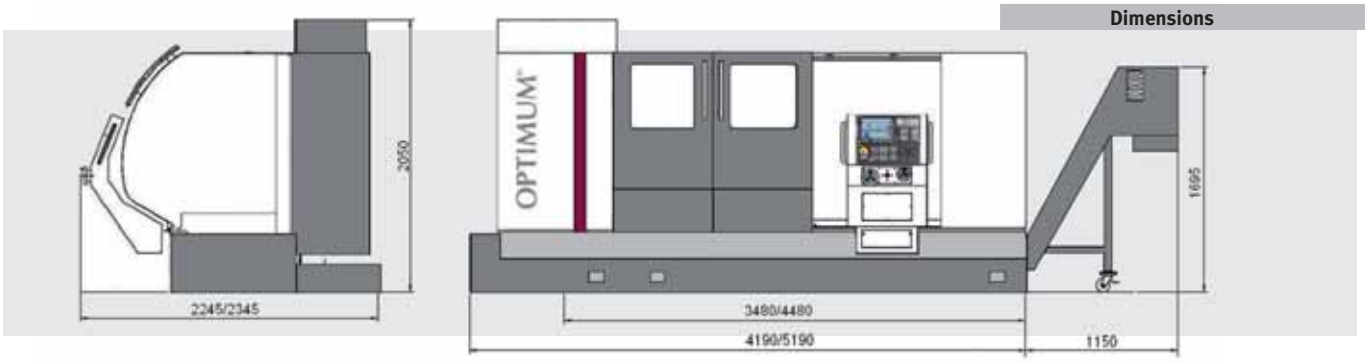
- A lifting device is required to unload the machine. The company Optimum may make it available by way of loan on bail in Germany and Austria. It may optionally be purchased.



„CNC-instruction - Information“ on page 80

„Starter kit VDI 40“ on page 75

Dimensions



Standard equipment

Machine bed



- Machine bed is designed particularly wide
- More rapid movement
- Ball screw drive for higher repeatability

Headstock



- incl. pneumatic automatic speed changer
- Fully automatic mode
- Shaft of the manual gearbox made of sintered bronze for high accuracy

Heat exchanger



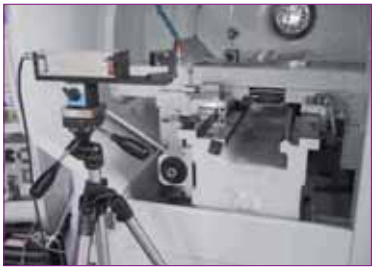
- Efficient integrated heat transfer from the switch cabinet
- Avoids that faults are generated from the overheated electrical and CNC components

Machine bed



- Machine bed designed for complete Z travel
- The actual turning capacity of the Z axis of other competitive machines on the market is 10-20% less efficient

Laser measurement



- Guaranteed repeatability and positioning accuracy

Turret



- 8 tool positions
- Hydraulic - VDI40
- Seat height max. 25 mm

Chip conveyor with chip carriage



- The chip conveyor eases the work and is time-saving

Chip carriage

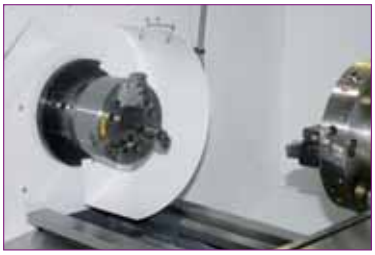
- Can be rolled and collapsed

Chip cleaning



- Easy cleaning of chips
- Inclined shape along the bed is available so that the chips directly fall on the chip tray
- The inclined design of the door avoids that the chips are getting caught in the door

Three-jaw chuck



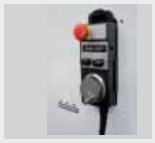
- Hydraulic chuck
- SizeØ 250 mm

Guide bearing



- Coated
- Manually scraped for highest accuracy

Special equipment

	Item No L 500	Item No L 520	Options
	351 4430 09	351 4440 09	**Non-rigid block jaws** for scope of delivery hydraulic 3-jaw chuck Ø 250 mm
	351 4430 06	351 4440 06	**3-jaw lathe chuck Ø 315 mm**
	351 4430 08	351 4440 08	**Non-rigid block jaws** for 3-jaw lathe chuck Ø 315 mm
	351 4430 20	351 4440 20	**3-jaw lathe chuck Ø 300 mm** (In exchange - Scope of delivery hydraulic 3-jaw chuck Ø 250 mm)
	351 4430 28	351 4440 26	**4-jaw lathe chuck hydraulic Ø 250 mm** (In exchange - Scope of delivery hydraulic 3-jaw chuck Ø 250 mm)
	351 4430 29	351 4440 27	**4-jaw lathe chuck hydraulic Ø 300 mm** (In exchange - Scope of delivery hydraulic 3-jaw chuck Ø 250 mm)
	351 4430 02	351 4440 02	**Joystick** to travel the X- and Z axis (not combinable with electronic handwheel)
	351 4430 17	351 4440 17	Detachable **electronic handwheel** (not combinable with joystick)
	351 4430 22	351 4440 22	**Oil separator**
	351 4430 11	351 4440 11	**Turret Baruffaldi TB 160 VDI40** 8 tools-driven tools (In exchange - Scope of delivery standard turret LS-160)
	351 4430 12	351 4440 12	**C axis brake system** (only with turret Baruffaldi TB 160)
	351 4430 10		**Drive motor 15 kw** (s1) (In exchange - Scope of delivery drive motor 12 kW)
	351 4430 21	351 4440 21	**Coolant pump** 5 bars
	351 4430 01	351 4440 01	**Coolant channels for tools 20 bars** (external)
	351 4430 27	351 4440 27	**Air cooling system** for switch cabinet (In exchange - Scope of delivery standard heat exchanger)
	351 4430 03	351 4440 03	**Steady rest** 20 mm - 200 mm
	351 4430 14	351 4440 14	**Steady rest** 150 mm - 330 mm
	351 4430 15	351 4440 15	**Steady rest** 180 mm - 410 mm
	351 4430 04	351 4440 04	**Follow rest** 20 mm - 200 mm
	351 4430 16	351 4440 16	**Preparation for hydraulic tailstock spindle**
	351 4430 18	351 4440 18	**Hydraulic tailstock spindle sleeve**
	351 4430 19	351 4440 19	**Extension of the tailstock sleeve** to 100 mm
	351 4430 31	351 4440 31	**Bar feeder 1,25 mtr.** V65-E-Pro
	351 4430 32	351 4440 32	**Bar feeder 1,5 mtr.** V65-LE-Pro
	351 4330 33	351 4410 33	**Bar feeder 1,25 mtr.** DH 65 Fedek
	351 4330 34	351 4410 34	**Bar feeder 1,5 mtr.** DH 65L Fedek

„CNC-instruction - Information“ on page 80

Can only be ordered ex works.
For prices and other options please contact your dealer or send us an e-mail: info@optimum-maschinen.de

Optimum CNC slant bed lathe: Speed, power, accuracy and durability

Convincing arguments: quality, efficiency and price

- Heavy type
- Compact type
- Slant bed type 30° for particularly large machining diameter
- Simple chip flow into the chip tray
- Dimensionally stable linear guiding - provides for long service life for maximum static and dynamic stiffness
- Hardened and ground ball screws
- SIEMENS servo motors for spindle, X- and Z axis
- All servo motors incl. integrated encoder for highest accuracy
- Equipment for manual tool measurement
- Tailstock with hydraulic spindle sleeve
- Electronic handwheel
- Two-year SIEMENS guaranty included

Set up with opened door

safety integrated power by siemens



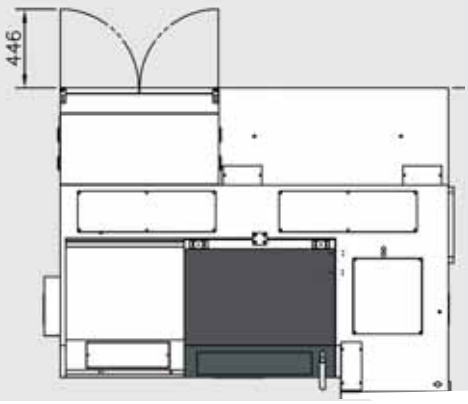
PREMIUM

Type	S 600
Item No	351 5060
Electrical connection	
Total connection	25 kW 400 V ~50 Hz
Driving motor	12 kW
Torque drive motor	115 Nm
Power of the coolant pump	700 W
Power of the cleaning pump	700 W
Tank capacity	140 litres
Hydraulic system	
Motor power	1.5 kW
Tank capacity	60 litres
Machine data	
Centre height	460 mm
Centre-Ø max.	210 mm
Turning Ø over cross slide	220 mm
Turning Ø over machine bed	500 mm
Inclined bed	30^0
Speeds	
Spindle speeds	10 - 4'000 min^{-1}
Spindle	
Spindle seat	DIN ISO 702-1 Nr. 6
Spindle bore	Ø 75 mm
Hydraulic lathe chuck	Ø 215 mm
Turret	
Type	hydraulic
Number of tool positions	12
Seat height, seat width max.	25 x 25 mm
Seat diameter max.	Ø 32 mm
Accuracy	
Repeatability	± 0.005 mm
Positioning accuracy	± 0.005 mm
Travels	
X axis	215 mm
Z axis	520 mm
Feed speed	
X-/Z-axis	30'000 mm/min
Torque motor	
X-/Z axis	11 Nm
Tailstock	
Tailstock seat	MT 4
Travel	370 mm
Tailstock - sleeve diameter	65 mm
Tailstock - sleeve travel hydraulically	50 mm
Dimensions	
Length x Width x Height	2'322 x 1'948 x 1'930 mm
Total weight	3'200 kg

Dimensions





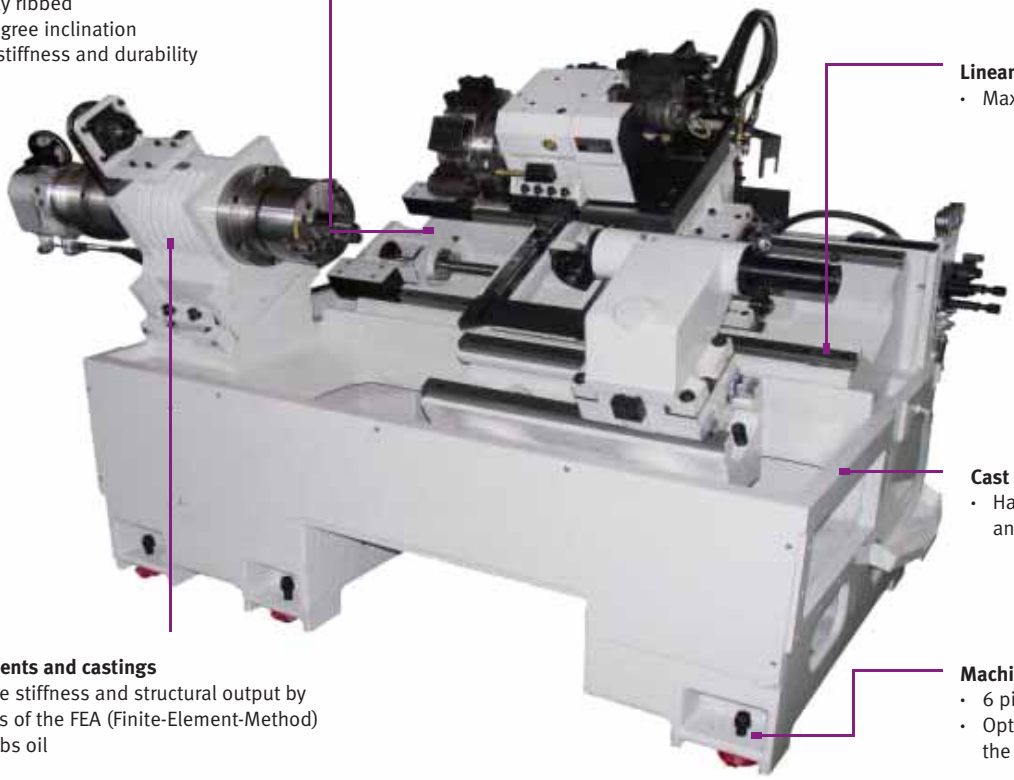


Standard equipment

Residual material detection and machining for contour pockets and stock removal	Hydraulic changer LS200 standard (12 tools)
Shop Turn	3-jaw chuck hydraulic Ø 215 mm
Network drive management	Heat exchanger
3-D simulation finished part	Chip conveyor
Simultaneous recording	Chip carriage
Safety integrated	Tool holder kit for LS 160
EMC	Coolant pistol
Manual handwheel	Tool holder for MT3
Automatic tool measuring probe Renishaw	Operating tool
Non-rigid and hard block jaws	
Toolholder kit (3 holders for reducing bushes, 1 holder for external lathe tool, 5 reducing bushes M12, M16, M20, M25, MT 3)	

Standard equipment

Machine bed
- Deeply ribbed
- 30 degree inclination
- High stiffness and durability



Linear guiding
- Maximum feed speed

Cast iron body
- Hardened (HRC48-52) and heat treated

Components and castings
- ensure stiffness and structural output by means of the FEA (Finite-Element-Method) absorbs oil

Machine feet
- 6 pieces
- Optimum alignment of the machine

Hydraulic tailstock spindle sleeve



- for a rapid processing it can be hydraulically run in and extended

Tool change system



- 12 tool positions
- Type of hydraulic turret
- Seat height max. 25 mm

Coolant and lubricant separator



- Separates the lubricant from the coolant

Hydraulic unit



- High capacity of the hydraulic oil tank
- Safety check valve
- Tank capacity 50 litres

Tool measuring probe



- Renishaw tool measuring probe allows measuring the tools within the machine

Central lubrication system



- Automatically
- Fold-out for easy filling
- The floating switch identifies the volume of the oil and releases an automatic alarm as soon as it falls below the set value

Special equipment

	Item No	Options
	351 5060 09	**Non-rigid block jaws** for for scope of delivery standard 3-jaw chuck Ø 215 mm
	351 5060 10	**Hard block jaws** for for scope of delivery standard 3-jaw chuck Ø 215 mm
	351 5060 08	**3-jaw lathe chuck Ø 250 mm** (In exchange - Scope of delivery hydraulic 3-jaw chuck Ø 215 mm)
	351 5060 24	**Hydraulic 4-jaw lathe chuck Ø 200 mm** (In exchange - Scope of delivery hydraulic 3-jaw chuck Ø 215 mm)
	351 5060 25	**Hydraulic 4-jaw lathe chuck Ø 250 mm** (In exchange - Scope of delivery hydraulic 3-jaw chuck Ø 215 mm)
	351 5060 20	**Oil separator**
	351 5060 01	**Turret type Sauter VDI 30** 12 tools - driven tools) and C axis brake system
	351 5060 22	**Coolant pump** 5 bars
	351 5060 19	**Coolant channels for tools** 20 bars external with external cooling tank
	351 5060 21	**Air cooling system** for switch cabinet (In exchange - Scope of delivery standard heat exchanger)
	351 5060 05	**Preparation for hydraulic tailstock spindle travel**
	351 5060 04	**Automatic tailstock travel per M code** (only in connection with a hydraulic tailstock spindle sleeve
	351 5060 03	**Tool changer VDI 30-40** In exchange - Scope of delivery standard turret LS-160
	351 5060 06	**Automatic tool measuring**
	351 5060 07	**Automatic part catcher**
	351 5060 11	**Bar feeder interface** for 351 5060 12 - 15
	351 5060 12	**Bar feeder 1.25 mtr.** V-65 E, PRO
	351 5060 13	**Bar feeder 1.50 mtr.** V-65 LE PRO
	351 5060 14	**Bar feeder 1.25 mtr.** DH65 FEDEK
	351 5060 15	**Bar feeder 1.50 mtr.** DH65 FEDEK
	351 5060 23	**Bar feeder 3.20 mtr.** for diameter 5-51 mm PRO
	351 5060 16	**Automatic door opening**
	351 5060 02	**Collet chuck for collets** from 15 - 60 mm
	351 5060 17	**Collet single** from 10 mm -14.9 mm
	351 5060 18	**Collet single** from 15 mm - 60 mm

Can only be ordered ex works.
For prices and further options
please contact your dealer or
send us an e-mail:
info@optimum-maschinen.de

Optimum CNC lathes - Higher speed, accuracy and efficiency with chip conveyor and C-axis

Convincing arguments: quality, efficiency and price

- Robust and heavy "Cartridge" spindle system incl. 1 two-row cylindrical roller bearing at the front and rear as well as double-sided taper bearing in the centre
- Long durability of all bearing due to permanent permanent lubrication
- All axes with SIEMENS servo motors
- Direct drive of all axis to eliminate the torsion play or for higher accuracy for thread cutting and contour machining
- Double pretensed ball bearing spindles with small pitch to increase the feed force
- Rapid turret indexing mechanism - releasing and turning process are practically taking place at the same time
- Turret indexing mechanism is performed non-stop bi-directionally
- Tailstock sleeve is activated by pressing the treadle or with the program
- Tailstock body is positioned by means of a control bar engaged by the slide
- Clamping and releasing of the drive bar can be programmed
- Electronic handwheel
- Bar feeder prepared
- Hydraulic turret
- Two-year SIEMENS guaranty included

SIEMENS SINUMERIK 828D

Siemens Safety Integrated

Set up with opened door

safety integrated power by siemens



PREMIUM

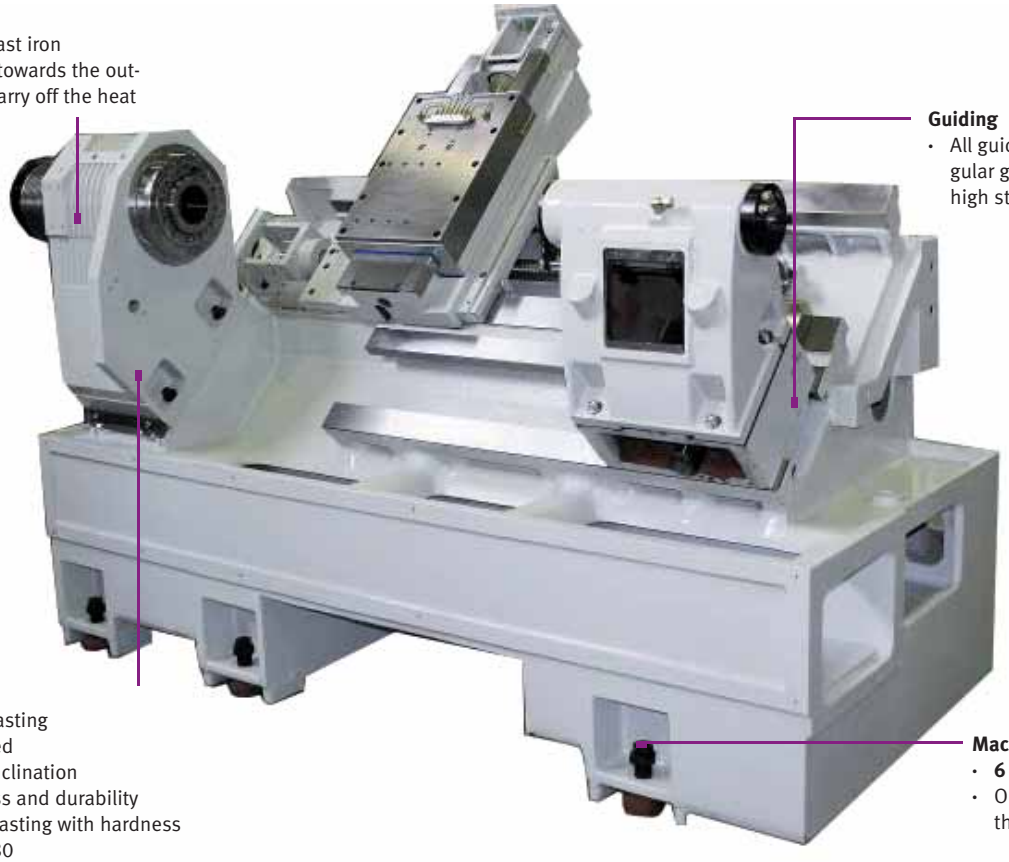
Standard equipment	
Residual material detection and machining for contour pockets and stock removal	Heat exchanger
Shop turn	Chip conveyor
Network drive management	Chip carriage
3-D simulation finished part	Automatic tool measuring Renishaw
Simultaneous recording	Automatic door opener
Safety integrated	Bar feeder interface
EMC	Coolant channels for tools 20 bars external
Working lamp	Driven tool changer VDI 40 (12 tools Sauter)
Manual handwheel	C axis brake system
Programmable tailstock	3-jaw chuck hydraulic Ø 200 mm
Parts catcher	

Type	S 500	S 750
Item No	351 5150	351 5170
Electrical connection		
Total connection	40 kW 400 V ~50 Hz	50 kW 400 V ~50 Hz
Driving motor	17 kW	30 kW
Torque drive motor	162 Nm	186 Nm
Power of the coolant pump	750 W	750 W
Tank capacity	185 litres	185 litres
Hydraulic system		
Motor power	1.5 kW	1.5 kW
Tank capacity	70 litres	70 litres
Machine data		
Centre height	600 mm	600 mm
Distance between centres max.	750 mm	1'250 mm
Turning Ø over cross slide	450 mm	450 mm
Turning Ø over machine bed	600 mm	600 mm
Turning diameter max.	485 mm	485 mm
Inclined bed	45^0	45^0
Speeds		
Spindle speeds	10 - 4'500 min^{-1}	10 - 3'000 min^{-1}
Spindle		
Spindle seat	DIN ISO 702-1 Nr. 6	DIN ISO 702-1 Nr. 8
Spindle bore	Ø 66 mm	Ø 88 mm
Hydraulic lathe chuck	Ø 200 mm	Ø 250 mm
Chuck through hole	Ø 52 mm	Ø 77 mm
Turret		
Seat	VDI 40	VDI 40
Number of tool positions	12	12
Max. speed of the tools	4'000 min^{-1}	4'000 min^{-1}
Power of the tools	4.82 KW	4.82 KW
Max. torque	20 Nm	20 Nm
Type	Sauter	Sauter
Seat height, seat width max.	25 x 25 mm	25 x 25 mm
Seat diameter max.	Ø 32 mm	Ø 32 mm
Accuracy		
Repeatability	± 0.005 mm	± 0.005 mm
Positioning accuracy	± 0.005 mm	± 0.005 mm
Travels		
X axis	305 mm	305 mm
Z axis	750 mm	1'250 mm
Feed speed		
X axis	24'000 mm/min	24'000 mm/min
Z axis	24'000 mm/min	24'000 mm/min
Torque motor		
X axis	11 Nm	11 Nm
Z axis	27 Nm	27 Nm
Tailstock		
Tailstock seat	MT 5	MT 5
Travel	600 mm	600 mm
Tailstock - sleeve diameter	90 mm	90 mm
Tailstock - sleeve travel hydraulically	120 mm	120 mm
Dimensions		
Length without / with chip conveyor	3'015 / 4'114 mm	3'515 / 4'614 mm
Width x Height	1'856 x 2'016 mm	1'856 x 2'016 mm
Total weight	5'700 kg	7'200 kg

„Starter kit VDI 40“ on page 75

Standard equipment

Headstock
- Meehinate cast iron
- Cooling ribs towards the outside better carry off the heat



Guiding
- All guiding devices are broad rectangular guiding devices and provide for high stiffness and stability

Machine bed
- One-piece casting
- Deeply ribbed
- 45 degree inclination
- High stiffness and durability
- Meehanite casting with hardness HB 170 ~ 180

Machine feet
- **6 pieces**
- Optimum alignment of the machine

Laser measurement



- Guaranteed repeatability and positioning accuracy

Tool change system



- High dividing precision and rapid tool change
- Slewing range, high torque and minimum stability
- Duration - tool to tool: 0.35 sec
- 180 degree: 1.2 sec

Programmable tailstock



- Heavy type
- High stiffness
- Tailstock can be activated via the program or directly controlled by the operator by means of a standard treadle

Internal tool cooling CTS



- Pressure: 20 bars
- Filtering accuracy 25µm

C axis brake system



- Hydraulic brake disc to check the effect of the C axis
- Spindle-servo-motor provides for higher resolution accuracy at 0.001°

Central lubrication system



- Automatically
- Fold-out for easy filling
- The floating switch identifies the volume of the oil and releases an automatic alarm as soon as it falls below the set value

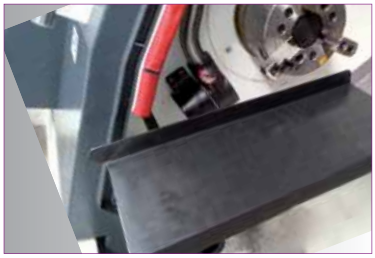
Special equipment

	Item No S 500	Item No S 750	Options
	351 5150 04	351 5170 04	**Non-rigid block jaws** for 3-jaw lathe chuck Ø 200 mm
	351 5150 05	351 5170 05	Non-rigid block jaws for 3-jaw lathe chuck Ø 250 mm
	351 5150 13	351 5170 13	**Hydraulic 4-jaw lathe chuck Ø 200 mm** (In exchange - Scope of delivery standard hydraulic 3-jaw lathe chuck Ø 200 mm)
	351 5150 16	-	**Hydraulic 3-jaw lathe chuck Ø 250 mm** (In exchange - Scope of delivery standard hydraulic 3-jaw lathe chuck Ø 200 mm)
	351 5150 17	-	**Hydraulic 4-jaw lathe chuck Ø 250 mm** (In exchange - Scope of delivery standard hydraulic 3-jaw lathe chuckØ 200 mm)
	-	351 5170 15	**Hydraulic 4-jaw lathe chuck Ø 300 mm** (In exchange - Scope of delivery standard hydraulic 3-jaw lathe chuck Ø 250 mm)
	351 5150 01	351 5170 01	**Oil separator**
	351 5150 03	351 5170 03	**Bar feeder 1.55 mtr.** V-65 LE, PRO
	-	351 5170 02	**Bar feeder 1.55 mtr.** DH-65 LE, FEDEK
	351 5150 12	351 5170 12	**Bar feeder 3.20 mtr.** for diameter 5 mm - 51 mm PRO
	351 5150 06	351 5170 06	**Axially driven tool holder**
	351 5150 07	351 5170 07	**Radially driven tool holder**
	351 5150 08	351 5170 08	**Rear radial driven tool holder**
	351 5150 09	351 5170 09	**Air cooling system** for switch cabinet
	351 5150 10	351 5170 10	**Steady rest** 20 mm - 200 mm
	351 5150 11	351 5170 11	**Tool changer VDI 40** (12 tools)

„CNC-instruction - Information“ on page 80

Can only be ordered ex works.
For prices and further options please contact your dealer or send us an e-mail:
info@optimum-maschinen.de

Part's catcher



- Automatic part catching device

Bar feeder prepared



- Connextion for bar feeder



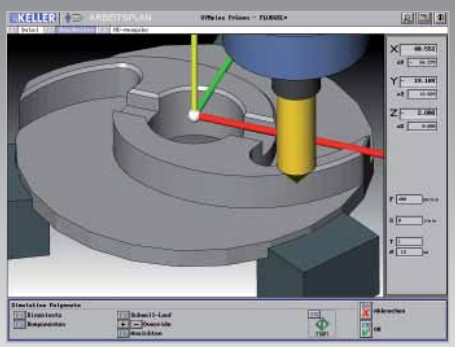
CNC software

SYMplus milling 60
SYMplus turning 62
Shopmill/ Shopturn 64
Sinutrain 65

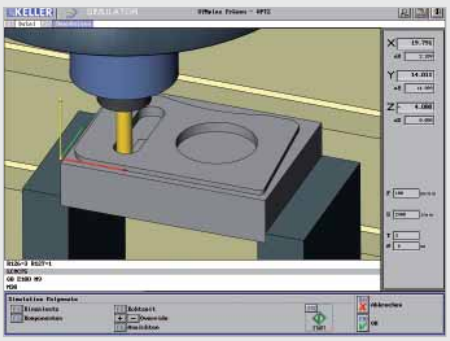
CNC controls

Sinumerik 802 S base line 66
Sinumerik 808D 68
Sinumerik 828D 70
Sinumerik 828D BASIC T 72

CNC software, controls and accessories







for cost-efficient, time-saving and economic working.

SYMplus milling is the perfect software add-on for all milling machines made by Optimum equipped with Siemens control.

CNC software SYMplus milling	358 3850
• Version 5.2	
Upgrade SYMplus milling	
For Siemens control **828 D**	358 3856
• Simulator and post processor	
For Siemens control **808 D**	**353 3859**
• Processor	
USB adapter	357 1968
• RS 232	

As training software it helps to rapidly introduce the operator how to use the control Sinumerik 802S. Employees having little CNC experience can learn the basics of the DIN programming by using SYMplus and are finally able to write and test programs using 802S cycles. In particular SYMplus is an easy-to-learn 2½D-CAD/CAM system which helps you to save programming times, to avoid crashes and to reduce the production time.

The software is subdivided into four modules:

SYMplus milling : CAD/CAM functionality at a glance

Geometry

- Interactive contour setting via pictograms. Any contour, rectangle, circle, polygon, drilling pattern, text
- Constructions of variants to simply change contours, bidirectional CAD interfaces (DXF, IGES)
- Measuring functions, 3D display at any point in time of the construction incl. calculation volumes and masses

Working plan

- Strategies reducing the production time by detection of residues over the whole processing, working with any raw parts
- Face milling, surface milling (pockets,islands), contour parallel or shaded with different plunging strategies (helical, ramp, vertical), contour milling, chamfering, grooving, engraving text, handwheel (teaching), clamping, drilling, sinking, grinding, threading
- Comprehensivee collision check, taking the clamping devices into account
- 2D and 3D simulation with common types of machines, real time, preview of single records, magnifying glass
- Time calculation for the calculation and to optimize the working plan, automatically created tooling sheet
- Post processor(s) to generate NC programs for your control

Tool management, data transfer

- Tool management (geometric and technological, depending on the material)
- NC editor for easy editing (copying, cutting, searching, replacing)
- Bi-directional data transfer via the serial interface (RS232)

SYMplus milling : Versions, licenses, delivery volume

- SYMplus Milling is available in the following languages: German, English, French, Spanish, Italian, Dutch, Polish, Hungarian and Slovakian. Other languages are available upon request.
- SYMplus will be delivered on CD-ROM. The license is protected by a USB dongle. The software package includes a manual and a 4-coloured exercise book of more than 130 pages.
- The software also always includes a SIEMENS post processor (matching the control of the F100 CNC resp. F 100 TC CNC) to automatically generate the NC programs for your machine. A simulator for the control SINUMERIK 802S and an RS232 interface to transfer the data is also integrated by default. Additional post processors and/or simulators are optionally available.

SYMplus milling : System requirements

- Commercial PC from e.g. Intel Pentium® IV on with min. 2 GHz
- Operating system Microsoft Windows® XP/Vista/7
- Screen resolution 1024x768
- OpenGL - compatible 3D graphic card, e.g. GeForce 7xxx/8xxx/9xxx (128 MB)
- Main storage: Microsoft Windows® 2000/XP 512 MB, Microsoft Windows® Vista/7 1 GB
- about 400 MB free hard disc storage for system data
- Sound card and loudspeakers

SYMplus milling : Module SHOP



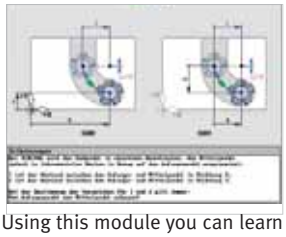
◀ Practice here typical repetitive sequences of action to set up the Sinumerik 802S.

◀ The realistically designed machine (with noises) will give you the feeling to practice very near to reality – without any stress of real risk of crash.

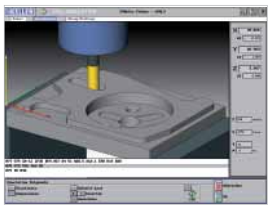
Also interactively practice the key sequence to program the Sinumerik control. ▶



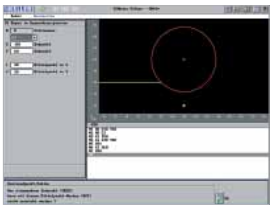
SYMplus milling : Module DIN/PAL



Using this module you can learn or freshen up the geometric basis of DIN-programming.

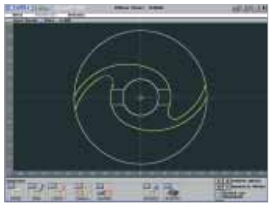


Complete DIN programs can be edited using a guided editor, checked for errorsand simulated.

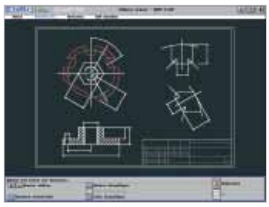


There is a context help and graphic support for error analysis. A test module is included for your trainees. The evaluation is performed automatically.

SYMplus milling : Module CAD/CAM - Geometry



Unprecedented in speed and ease you can graphically program workpieces using SYMplus even if the drawing is not dimensioned according to the NC dimensioning.



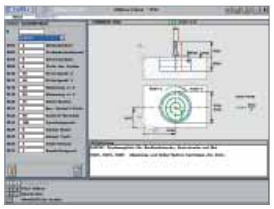
Alternatively you can load and output CAD contours.

Value entries are immediately dynamically converted.



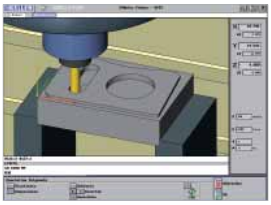
It is also possible to subsequently change any construction data within short time. It is possible to create symmetric contours by mirroring/turning, etc. Also measurement geometrics can be created with a few mouse clicks.

SYMplus milling : Module CONTROL

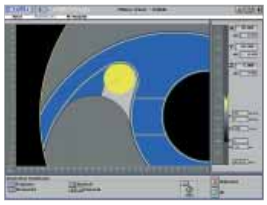


◀ Help graphics are available for all commands and cycles. The clear editor allows you to work efficiently.

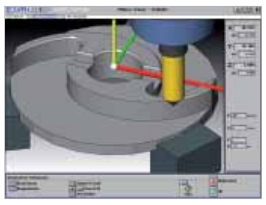
Here you can write and simulate programs using 802S cycles and commands. ▶



YMplus milling : Module CAD/CAM - Working plan

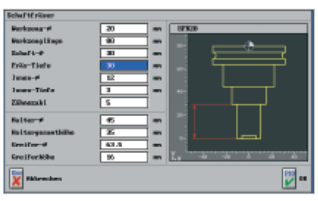


◀ The 2D simulation shows lots of details in the view from the top such as e.g. measurements and the cutting trace of each single milling path.



◀ The 3D simulation will give you the best overview of the operations. Also chamfers and depressions are easy to recognize.

SYMplus milling : Setting up tools, etc.



◀ Tools can be entered geometrically including shank and holder – this is important for the crash control.

The tools can be individually configured in magazines..



CAD/CAM system including 802S training appropriate for shops.

SYMplus turning is the ideal software supplement for CNC lathes made by OPTIMUM.

CNC software SYMplus turning	358 3852
• For Siemens control 802S base line	
• Version 5.2	

Upgrade SYMplus turning	
For Siemens control **828 D**	358 3854
• Simulator and post processor	
For Siemens control **808 D**	**353 3858**
• Post processor	
USB adapter for Siemens	357 1968
• RS 232	

As training software it helps for rapid introduction in the operation of the control Sinumerik 802S. Employees having little CNC experience can learn the basics of the DIN programming by using SYMplus and are finally able to write and test programs using 802S cycles.

In particular SYMplus is an easy-to-learn CAD/CAM system which helps you to save programming times, to avoid crashes and to reduce the production time.

The software is subdivided into four modules:

SYMplus turning : CAD/CAM functionality at a glance

Geometry

- Interactive contour setting via pictograms, any contour routings, cylinder, stage pins, tubes, grooves,
- Constructions of variants to simply change contours, bidirectional CAD interfaces (DXF, IGES)
- Measuring functions, 3D display at any point in time of the construction incl. calculation volumes and masses

Working plan

- Strategies reducing the production time by detection of residues over the whole processing, working with any raw parts
- Roughing plane and transversal finishing, engraving, punch roughing, cutting-off, drilling, threading, thread turning, handwheel (teaching), re-chucking
- Comprehensive collision control
- 2D and 3D simulation, real-time, single block preview, magnifying glass
- Time calculation for the calculation and to optimize the working plan, automatically created tooling sheet
- Post processor(s) to generate NC programs for your control

Tool management, data transfer

- Tool management (geometric and technological, depending on the material)
- NC editor for easy editing (copying, cutting, searching, replacing)
- Bi-directional data transfer via the serial interface (RS232)

Versions, licenses, delivery volume

- SYMplus turning is available in the languages German and English. Other languages are available upon request.
- SYMplus will be delivered on CD-ROM. The license is protected by a USB dongle. The software package includes a manual and a 4-coloured exercise book of more than 130 pages.
- The software also always includes a SIEMENS post processor (matching the control of the Optimum machines) to automatically generate the NC programs of your machine. A simulator for the control SINUMERIK 802S and an RS232 interface to transfer the data is also integrated by default. Additional post processors and/or simulators are optionally available.

System requirements

- Commercial PC from e.g. Intel Pentium® IV on with min. 2 GHz
- Operating system Microsoft Windows® XP/Vista/7
- Screen resolution 1024x768
- OpenGL - compatible 3D graphic card, e.g. GeForce 7xxx/8xxx/9xxx (128 MB)
- Main storage: Microsoft Windows® 2000/XP 512 MB, Microsoft Windows® Vista/7 1 GB
- about 400 MB free hard disc storage for system data

SYMplus turning : Module SHOP



◄ Practice here typical repetitive sequences of action to set up the Sinumerik 802S.

◄ The realistically designed machine (with noises) will give you the feeling to practice very near to reality – without any stress of real risk of crash.

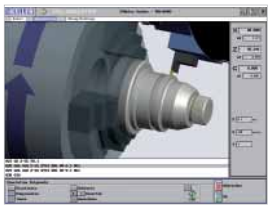
Also interactively practice the key sequence toprogram the Sinumerik control. ►



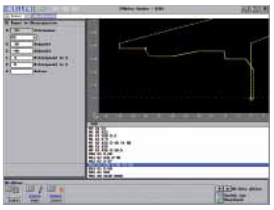
SYMplus turning : Module DIN/PAL



In this module it is possible to learn or freshen up geometric basis of the DIN/PAL programming.

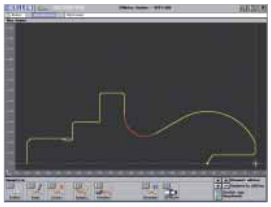


Complete DIN/PAL programs can be edited using a guided editor, checked on errors and simulated.

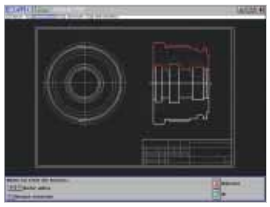


There is a context help and graphic support for error analysis.

SYMplus turning : Module CAD/CAM - Geometry

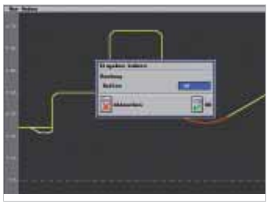


Unprecedented in speed and ease you can graphically program workpieces using SYMplus even if the drawing is not dimensioned according to the NC dimensioning.



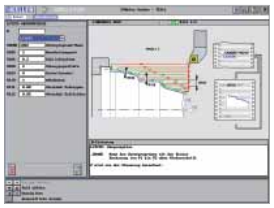
Alternatively you can load and output CAD contours.

Value entries are immediately dynamically converted.



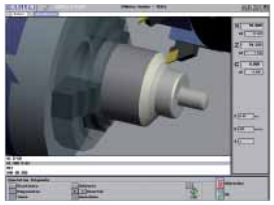
It is also possible to subsequently change any construction data within short time. Fit sizes (about 48h7) can be directly entered, average tolerance is automatically proposed as nominal dimension

SYMplus turning : Module CONTROL

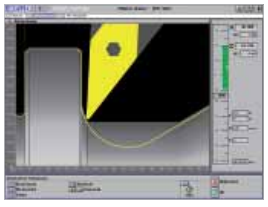


◄Help graphics are available for all commands and cycles. The clearly arranged editor allows you working efficiently.

Here you can write and simulate programs using 802S cycles and commands. ►



SYMplus turning : Module CAD/CAM - Working plan

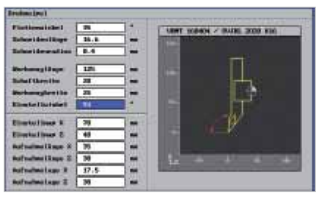


◄ The 2D simulation shows lots of details such as e.g. allowances. Here it is also possible to measure the workpiece



◄ The 3D simulation will give you the best overview of the operations.

SYMplus turning : Setting up tools, etc.

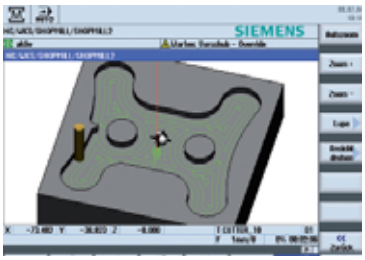


◄ Tools can be geometrically entered incl. setting angle, shank and seat – this is important for the crash control.

Finally the turret is equipped. Each equipment can be individually saved.. ►



ShopMill is a tailored technology package for all standard CNC milling machines



The main focus of these machines is on the production of individual parts and small series. ShopMill offers a simple operation of the machine tool appropriate for skilled workers.
All operating actions are supported by help graphics.
The functions in the manual mode allow rapid, practice-oriented setting-up of the machine for the operation. Those are in particular determining the workpiece position in the machine as well as the maintenance and measurement of the used tools.

For the programming, ShopMill is offering two different programming tools.
- The DIN/ISO editor serves to create DIN/ISO programs on the machine as well as to correct externally created DIN/ ISO programs.
- The working plan editor serves for graphic programming on the machine.

Thus, ShopMill offers a uniform control configuration which covers all worldwide required fields of application without subsequent commissioning efforts:
- Simple user interface for all machine functions
- DIN/ISO programming on the machine or offline via CAD/CAM system
- Graphic programming
- Mould making applications
- Measurement functions

ShopMill - Highlights
- Flexible programming by graphic interactive programming inputs (without DIN/ISO knowledge) or textual programming inputs with practical cycles also for the use in the tool and mould making
- User-friendly operation by practice-oriented setting-up and measuring functions, clearly arranged tool management and 3D simulation
- Increase in productivity by supporting PC software for production planning without machine scheduling

ShopTurn - An operation and programming software - distinguishes itself by a graphically supported programming and working plan creation

In ShopTurn machines are rapidly set up and workpieces and tools are rapidly measured. The operating and programming software ShopTurn distinguishes itself by a graphically supported programming and working schedule preparation, by means of which it is possible to manufacture a workpiece even without knowledge of DIN/ISO programming commands.



It is possible to call a graphic overall representation of the workpiece from the working plan by a push of a button. With the help of this dynamic line graphic it is possible to rapidly identify and correct programming errors.
Special advantages: By means of simulation it is possible to test the created working plan. During the simulation run, the productive time of the turned part is determined simultaneously. In this way, it is possible to program and simulate the workpiece in short time after the inquiry from the drawing using ShopTurn and in addition to create a price statement starting from the determined productive time: time saving for the customers.

ShopTurn - Highlights

The following basic elements are available:
- Straight line X/Z also with C axis
- Incline in X/Z or under angle, also with C axis
- Circle parametrizable by radius / endpoint or centre

Manual functions/Set up functions
Measuring workpiece › Scratching workpiece in Z axis
Measuring workpiece › Measuring tool using Tooleye
Measuring tool by scratching

Other options
Easy-to-understand help graphics for each cycle
Scale drawing graphic for input support
Graphic working plan test / test run
Creating / reading and processing of complete DIN/ISO programs
Technology input workpiece definition for each working plan step is possible
Tool and wearing data for up to 128 tools

Automatic functions
Block scan (also on individual holes of a drilling template)
Sequence (automatic operation)
Working plan test (Dry-Run)
Restarting at the contour
Block search
Other functions
The step-by-step creation of a working plan allows you to characterize each step by easy-to-understand icons, working plan creation without DIN/ISO knowledge simple modifying, inserting and removing of special commands (e.g. M functions) 8 inserting of transition elements, step-by-step processing (single step)

NEW

OPTIMUM® MASCHINEN - GERMANY

SIEMENS SINUTRAIN - SINUMERIK training software

Software	Item No
Siemens Shopturn	900 1030
Siemens Shopmill	900 1031

Software	Item No
Siemens Shopturn & Shopmill	900 1032
Sinumerik 808D on PC free download at cnc4you	

Learning technique, controlling technique: Formation and training with SINUTRAIN

SinuTrain is the control identical software of SINUMERIK for your PC. From training courses via the offline programming up to the professional sales presentations – SinuTrain is designed for the different requirements.
With SinuTrain we are offering you a practice-oriented and integrated solution for CNC training courses. From the basics to the professional qualification you can
easily mediate your function knowledge of the innovative CNC control SINUMERIK using this training software. Our holistically oriented training solution perfectly includes a training software
and systems which are perfectly adapted to your requirements.

The advantages at a glance

- Comfortable SINUMERIK operation and CNC programming on your PC. Suitable for all common CNC programming types.
- Optimum machine control via the machine operating panel which is integrated in the software.
- Optimum adaptation to different machines
- in the sense of maximum compatibility of the
- CNC programs. Available in lots of languages.



SinuTrain – more than a training tool.

The operation of SinuTrain is identical with the operation of the real SINUMERIK system. Accordingly
you may test all operating modes of the user interface SINUMERIK Operate step by step. Beside milling and turning applications SinuTrain can also be used for other technologies on the basis of SINUMERIK Operate.

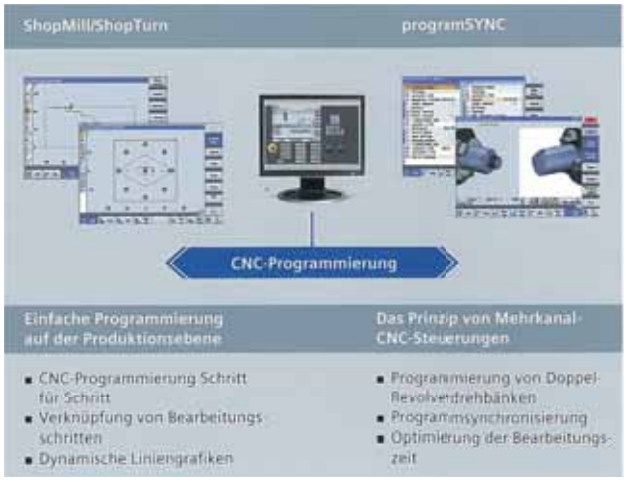
As real as reality

SinuTrain allows a comfortable SINUMERIK operation and CNC programming on your PC – on the basis of the real SINUMERIK CNC kernel.
The software can perfectly be adapted to the axis configuration of different machines. In this way, maximum compatibility of offline created CNC programs is ensured with the machines in the production area.

SinuTrain – numerous useful additional functions

SinuTrain offers numerous useful additional functions and proves your performance during operation.
SinuTrain allows a comfortable SINUMERIK operation
and CNC programming on your PC – on the basis of the real SINUMERIK CNC kernel. The software can perfectly be adapted to the axis configuration of different machines. In this way maximum compatibility of offline created CNC programs is ensured with the machines in the production area.

- The integrated online help for CNC operation and programming is making the correct information available.
- An integrated manual facilitates the approach to CNC programming.
- Printout of DIN/ISO programs and working step programs provide an overview.
- Parallel installation of different versions of the SinuTrain software can be easily performed.
- Windows application facilitates the installation
- Communicative by program transfer via Ethernet or USB to your machine



Please find further information under:
info@optimum-maschinen.de

Sinumerik 802 S base line

The ideal CNC for standard applications

SINUMERIK 802S base line, the CNC made by Siemens for applications in the low power range is excellently appropriate for lathes and milling machines with up to 3 axis and one spindle and distinguishes itself by the following features of:

- Extremely easy-to-operate and simple programming
- Maintenance-free
- Compact design

For Optimum machines
M4HS
L 28
L 33



The perfect control for starting:

- Easy-to-operate and to program
- Rapid, easy setting-up of tools and workpieces
- Cycle and contour-supported programming

The CNC control for different applications:

- Controls up to 3 axes and 1 spindle
- Complete unit of compact dimensions
- Maintenance-free, since it is working without batteries and fans
- Easy connection by pulse/direction interfaces (feed axes) resp. ± 10 V analogue interface (spindle)
- Standard solution with worldwide service for different applications

The SINUMERIK 802S base line offers you lots of advantages:

- Easy CNC programming in DIN 66025
- Easy measuring of tools by scratching
- Contour elements as programming aid
- Cycle support for complex applications
- Spindle pitch error, batch and measuring system error compensation for precise processing results.

Turning cycles:

- Grooving
- Undercut
- Machining
- Thread cutting*
- Drilling, spot facing*
- Deep hole drilling*
- Tapping with/without compensation chuck*
- Drilling

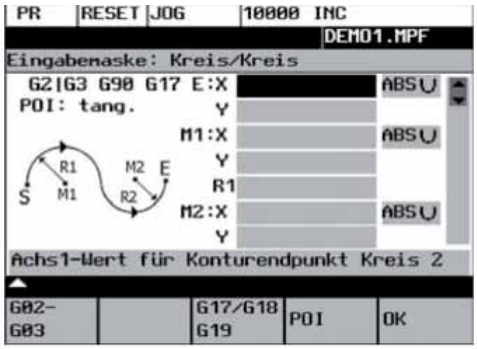
Milling cycles:

- Deep-hole drilling
- Thread cutting with/without compensation chuck
- Row of holes – circle
- Row of holes – line
- Drilling, countersinking
- Milling of pockets, grooves and circular pockets

Other features:

- Processing from external – also processes larger CNC programs via the RS 232 C (V.24) interface
- Extremely robust – due to compact design
- Powerful PLC – for perfect adaptation to the machine tool (SIMATIC S7-200 compatible)
- 256kb program memory

*depending on the machine type

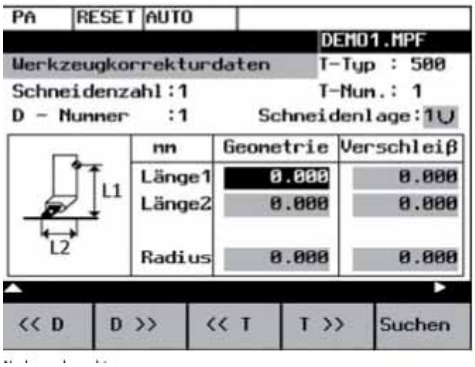


Programming

- Program entry according to DIN 66025
- High level language elements and R parameters for modular and efficient programming
- Graphic programming by means of contour routing support
- Integrated turning and milling cycles with graphic input masks
- Teach in (Optional) for program entry by means of handwheels or direction keys

Tool management

- 15 tools and 30 cutting edges
- Tool length compensation
- 2D tool radius compensation
- Measuring cycles for easy workpiece and tool measurement



Tool corrections

Dimensions

- 3 servo axis (pulse/direction interface, 5V difference signals) and 1 spindle (±10 V)
- 8" LCD flat screen (monochrome)
- 256 kbyte part program memory
- Handwheel function (max. 2 handwheels)
- Serial interface RS 232 C
- Operating panel front of IP65 type
- 12 customer keys with LED can be freely allocated and determined
- 48 digital inputs and 16 digital outputs are integrated
- Extension module with 16 digital inputs and outputs each (max. 2 pieces)
- Emergency-stop button can be integrated
- PLC with contact plan programming
- Look Ahead function
- Up to 15 tools, 30 cutting edges
- Calculator function
- Servo trace for axis diagnosis
- 2 languages can be switched over online
- Operating software with 11 languages (English, German,French, Italian, Spanish, Chinese simplified, Polish, Russian, Czech, Turkish and Hungarian)

SINUMERIK 808D

relating a high degree of user friendliness with high performance CNC functions.

The CNC SINUMERIK 808D Milling is perfectly pre-configured for the requirements of easy standard milling machines. The SINUMERIK 808D Milling is equipped with the intelligent move guiding advanced surface of the technology package SINUMERIK MDynamics. It also includes the LookAhead function which allows an optimum speed control.

- **SINUMERIK Operate BASIC**
  easy, intuitive user guiding by dialogue-oriented user support
- **SINUMERIK programGUIDE BASIC**
  Includes a bandwidth of technology cycles for milling and drilling with graphic input masks
  High performance and accuracy thanks to modern CNC functions
- **SINUMERIK MDynamics**
  Intelligent move guiding advanced surface for applications in the simple tool making and mould making
- **Machine Control Panel**
  The machine control panel can be connected as "Plug&Play" via USB and is equipped with ergonomic override rotary switches in the same way as the High-End-CNCs.

The compact and user friendly entry-level solution is used for simple milling applications. As panel-based control the 808D is equipped with all interfaces including PLC in- and outputs on board so that no complex cabling to the switch cabinet is required any longer. The machine control panel can be connected as "Plug&Play" via USB and is equipped with ergonomic override rotary switches in the same way as the High-End-CNCs. The interface on the operating panel front allows the use of a USB memory stick and thus the easy transmission of part programs and tool data during the daily operation.

A significant feature of the new CNC is the easy, intuitive user guidance. The user interface Sinumerik Operate Basic offers a new dialogue-oriented user support with the Sinumerik startGuide which is guiding through the setting of the machine step-by-step and reduces the time for commission-

For Optimum machines

M2LS
F 4
F 80
F 105



ing to a minimum. Thus it is possible to easily commission prototypes as well as serial machines by means of the so called "startupAssistants". The "operationAssistant" helps the operators to set-up and program the machine step by step. All in all the completely renewed Sinumerik Operate Basic is modelled strongly on the operating philosophy of the larger Sinumerik Operate system. This applies for the menu structure as well as for the programming. In this way the "programGuide Basic" allows easy programming by means of graphically supported cycle masks and a comfortable contour calculator. The reuse of part programs of other controls which support the ISO code can be easily realized. In this way it is possible to use DIN- and ISO programming languages alternately within a program. As usual, the user interface offers an online language changeover. On the hardware key labels for the CNC are available with English and Chinese characters. Thanks to the used micro keys the membrane keyboard is also offering a very good tactility.

User friendly:
Technology specific
keyboard layout
Hardkeys with protective film

Display:
7.5" colour LCD display
with
a resolution of 640 x 480

Communicative:
USB interface at the front
panel supports USB memory
stick and USB PC keyboard



Robust:
Panel-based CNC
design
Protection class
IP65 - at the front

Easy to assemble and to maintain:
Durable buffer battery
Clamping assembly of
the PPU and MCP

Easy-to-operate:
MCP with rotary switch for
feed and spindle override
LED display of the tool number

The new model shows a high degree of precision and productivity. Thus the CNC is counting with an accuracy of 80-Bit-NanoFP in order to avoid internal rounding errors. Being the smallest member of the Sinumerik family the 808D is equipped with the intelligent move guiding MDynamics including an intelligent"Look-Ahead" to attain an optimum speed guiding. Thus it is possible to attain productivity increases and improved service quality at the same time. The high computing power as well as optimized technology cycles in the operating system contribute to the high performance of the CNC. The transfer from conventional machines to CNC programming is enabled by using the option Manual Machine plus (MM+). A machine which is equipped with this feature can be operated by means of handwheels in the same way as a conventional machine but with the advantages of CNC supported technology cycles.

At the driving end the control is equipped with a semi-servo solution consisting of the drive system Sinamics V60 combined with Simotics servo motors of the type 1FL5. The CNC provides the drive with target values via the plus-/direction interface which the CNC perfectly translates in the enclosed speed control circuit by means of the Simotics motor. The continuous CNC system can also be used in rough environmental conditions among others since the drive system is equipped with a large cooling body which can be operated without ventilator. Furthermore the Sinumerik 808D attains the protection class IP65 on the operating panel front.

The new CNC is completed with the software package Sinumerik 808D on PC which further facilitates the use of the machine. When used as practical training software it is possible to program and simulate workpieces offline. The free of charge software can also be used for presentation purposes.

Language Note:
Version 1 of the following languages: D / GB / RUS / P / CN
Version 2: languages on request

Sinumerik 828D

High performance CNC control for maximum accuracy and processing speed

SINUMERIK 828D is a panel-based CNC control for demanding applications on lathes and milling machines as they are typically used in the shop. They combine CNC, PLC, operating and axis control functions on a compact and robust unit which fits in each operation panel housing. There are two variants for the horizontal and vertical installation.

With its powerful CNC functions the SINUMERIK 828D sets new benchmarks in the class of compact CNCs. Thanks to a comprehensive CNC programming package the Sinumerik 828D is perfectly prepared for all worldwide CNC consumption markets.

For Optimum machines

F 150
F 310
F 410
S 500
S 750



Compact
Maximum power with most compact dimensions:

- 10.4" colour display
- QWERTY CNC standard keyboard
- USB, CF card, Ethernet at the panel front side

Powerful
High performance CNC functions

- 80bit NANOFP accuracy
- Technology package SINUMERIK MDynamics with the new function Advanced Surface
- Clearly arranged tool management
- Powerful kinematic transformations

Simple
Simple programming

- ShopMill / ShopTurn - shortest programming times for individual parts and small series
- programGUIDE- shortest program runtimes and maximum flexibility for large batch sizes
- ISO dialect – optimum CNC program compatibility

Simple

- Easy commissioning
- Easy Archive – optimum application updates
- Easy Extend - user-friendly extension of machine components

Simply brilliant

- Animated elements - unique graphic visualisation
- Integrated online help - rapid relevant help
- Easy Message – Transfer of the machine status by SMS

Scope of service	
Minimum block cycle time	~6 ms
CNC main storage	3 MB
Maximum number of tools/cutters	128/256
Maximum number of axes/spindles	6

CNC operation in the manual mode (JOG)



TSM universal cycle

A universal cycle is available in the setting mode for the most frequently used machine functions:
- Tool change with direct access via the tool table (T)
- Spindle speed and –direction (S)
- M functions (M)
- Activating of zero offsets



Turning cycle

In the setting mode a comfortable turning cycle is available. Using this cycle it is for instance possible to turn smooth clamping jaws.
It is possible to preset the following parameters:
- Roughing resp. finishing
- Undercut for smooth clamping jaws



Measuring workpiece

You can measure the workpiece as follows:
- Reference tool

Zero point offsets

The following adjustable zero point offset are at your disposal:
- A basic offset
- Maximum 99 zero point offsets (G54, G55 ...)
- Each zero point offset with axis rotation and fine offset

Tool management

Tool table



You can manage tools including their complete use data in the tool list.
The maximum number of tools/cutters amounts to 128/256
- The function loading allows to assign the tools to the required magazine positions.
- You can store the following data for each tool:

– Tool type: e.g. roughing, finishing, engraving, drilling and milling tools
– Clear tool name in clear text, example: ROUGHING_80GRAD
– Max. 9 cutters for each tool
– Tool lengths and cutting insert geometry
– Point angle for drilling and number of teeth for milling tools
– Spindle turning direction and coolant (Step 1 and 2)

- Direct taking over of the tool from the list to the program resp. to measurements

Monitoring service life and number of pieces

Using the SINUMERIK 828D you can monitor the service life of your tools and the number of substitutions. You can assign clear names to your tools instead of less significant numbers. At the least when reading the CNC program you will appreciate this comfort.
- Monitoring contact time (T) in minutes or number of exchanges (C)
- Early warning threshold for making available new tools in time
- If the required tool is not in the magazine the SINUMERIK 828D prompts you to change it manually.

Spare tools

If required you can also manage spare tools (replacement tools) with the SINUMERIK 828D. Tools of the same name are created as spare tools. In the column ST the spare tools are marked with an ascending number.

Sinumerik 828D BASIC T

The all-round talent for standard lathes

The SINUMERIK 828D BASIC T is the perfect entry into the class 828D and together with SINAMICS drives and motors precisely tailored for the demands of modern standard lathes.

The SINUMERIK 828D BASIC T controls all drilling and milling operations besides numerous drilling operations. Of course, also on the fore-part and surface area of the workpiece. t this, the one-time performance of the SINUMERIK 828D BASIC T guarantees highest accuracy of the workpiece with minimum processing time.

For Optimum machines

L 44
L 440
L 460
L 500
L 520
S 600



Robust and maintenance-friendly Sinamics drive

In addition to new CNC Siemens is offering a multiple axis drive with the new Sinamics S120 Combi which is tailored to the demands of compact standard machines in the same way as the CNC controls. Robustness is an important feature for the Combi drive. In order to guarantee this among others the supply unit as well as the three to four motor modules are integrated in one housing. In this way, on the one hand the number of cabling interfaces is decreased, on the other hand a mutual cooling element provides for optimum heat dissipation – even if one of the performance parts is under full load. If the machines are used in tropical environmental conditions, the drives have to withstand tension and frequency variations in the power network without malfunctions beside heat resistance and condensation tightness. The Sinamics S120 Combi was also designed for these conditions.

The Sinumerik 828D Basic T is already integrated in the training package SinuTrain for Sinumerik Operate. In this way, it is possible to learn and deepen the operation of a lathe with the latest control. Furthermore, SinuTrain is also appropriate for performing the programming offline. At this, it is possible to use the established programming methods Sinumerik-CNC programming language, ISO-Code programming as well as the step sequence programming ShopTurn in the same way as on the real CNC in the machine.

With this type Siemens Drive Technologies complements its portfolio of controls in the compact class Sinumerik 828D. With the Sinumerik 828D for demanding lathes and milling machines in the workshop , the Sinumerik 840D sl for premium applications and the Sinumerik 802 family for standard machine applications make available a complete spectrum of machine tool controls for all CNC requirements.

High-tech for the compact class

Modern processor technique and software architecture are the basis of the 80-bit NANOFP accuracy (80 Bit floating point precision). From the software point of view it is possible to attain accuracy values in the range of microns (µm). Thus the accuracy which can be attained practice is not limited control technically , but it is determined by the options of the mechanical components. Beside the high accuracy the Sinumerik 828D Basic T furthermore allows very flexible use since it can also perform drilling and milling processes beside numerous turning operations. With intelligent kinematic transformations between the spindle and the X-axis it is even possible to perform different milling and drilling operations on the face and lateral surfaces of the workpiece. The Sinumerik 828D Basic T is equipped with Sinumerik Operate, a comfortable CNC user interface which allows performing safe and intuitive programming and setting up processes.

The function "Easy Message" is a special feature. In this way, the responsible employee is informed via SMS about all important process data regarding production. If a machine requires new raw parts or a lathe tool attained the limits of its lifetime the Sinumerik 828D Basic T can send a corresponding information via SMS. In this way, the productivity of the machine is further increased.

Scope of service	
Input voltage	DC 24 V, + 20 %/- 15%
Power consumption, max.	60 W
Network failure bridging operation	3 ms
Protection class according to DIN EN 60529 (IEC 60529)	Operating panel front IP65 (with closed front cover)
	PPU IP20 (rear)

Overview

Compact, strong, simple - just brilliant
The compact, operating panel based CNC control SINUMERIK 828D BASIC T is offering a maximum of robustness and maintenance-friendliness.
Powerful CNC functions coupled with a unique 80-bit NANOFP accuracy providing highest workpiece accuracy with minimum operating time. Thanks to a flexible CNC programming language as well as a unique working step programming ShopTurn/ ShopMill, it is possible to program and edit parts for mass production as well as individual components with the highest possible efficiency.

Preconfigured technology-specific system software and unique service functions reduce the commissioning costs and the service to an absolute minimum.

Advantage

- Compact, robust and maintenance-free operating panel CNC with firm system software for the technology turning
- Highest operating precision with 80-bit NANOFP accuracy
- New user interface SINUMERIK Operate, continuous for SINUMERIK 840D sl
- Intelligent kinematic transformations for the milling and drilling operations on the front and lateral surfaces of the workpieces
- ShopTurn: shortest programming time when producing individual parts and small series
- programGUIDE: shortest operating time and maximum flexibility when producing parts for mass production
- Unique spectrum of technology cycles – from the operation of any turning and milling contours with detection of residual material up to process measurement
- Animated elements: unique operating and programming support with moving image sequences
- Most modern data transfer via CompactFlash Card, USB stick and factory network (Ethernet)
- Easy Message: maximum machine availability by process monitoring via text messages (SMS)

Function

- 2 operating panel types for horizontal and vertical operating panel housings
- Integrated QWERTY-CNC standard keypad with short-stroke keys
- CompactFlash Card, USB and Ethernet interface on the operating panel front
- Additional Ethernet interface on the CNC rear side for firm factory networking
- Integrated PLC on the basis of the instruction set of the SIMATIC S7-200 with ladder diagram programming (Ladder Steps)
- E/A interface is based on PROFINET for the connection of the PLC peripherals and machine control panel
- Licensing CNC options
- Up to 5 axis/spindles
- 1 processing channel/group of operating modes
- Integrated tool management with tool lifetime control
- Management of spare tools (optional)
- User images can be projected Easy Screen
- Integrated data archiving procedure for simple data update
- Provision of repair of deficiencies for 24 months from the 2nd commissioning on for all equipment components according to the RSV work description

Integration

It is possible to connect the following components to the SINUMERIK 828D
BASIC T:

- Up to 2 electronic handwheels
- Mini handheld unit with handwheel
- Up to 3 peripheral modules PP 72/48D PN or PP 72/48D 2/2A PN
- Machine control panel MCP 310C PN or MCP 483C PN
- GSM/GPRS modem
- Drive system SINAMICS S120 Combi via DRIVE-CLiQ

Safety engineering - Safety integrated



SINUMERIK Safety Integrated ®
is the comprehensive safety pack for person and machine protection, is working extremely reliable, efficient and profitable. It guarantees safe and practicable operation of the machine for all required operating conditions. All safety functions fulfil the requirements according to SL 2 /PL d and are certified accordingly and listed under NRTL.
Comprehensive, highly active safety functions result in rapid, situation-related reactions in case of a fault due to the integration of the safety functions in the drive and control. Among others, the safe monitoring of speed, standstill and position are integrated.
The safe brake management allows anti-fall guard for vertical axis. The security-relevant signals can be linked due to the safe, programmable logics without any additional hardware.

Starter kit VDI 30

Starter kit	VDI 30
Item No	353 6115

Consisting of:

- 3 pcs Square transversal seat
- 1 pcs Square transversal seat overhead
- 1 pcs Longitudinal seat
- 5 pcs Bohrstangenaufnahme Ø 10 / 12 / 16 / 20 / 25 mm
- 3 pcs Cover plate
- 1 pcs Collet chuck ER 25
- 1 pcs Key for collet chuck ER 25
- 1 pcs Collet chuck kit ER 25
- 1 pcs Tool holder
- 1 pcs Bohrfutter

Square transversal seat 353 6231

- Right, short
- DIN 69880
- large adjustable ball-type nozzles



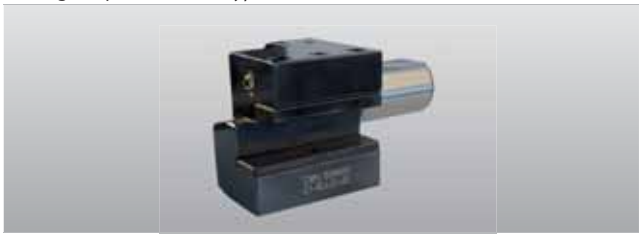
Square transversal seat 353 6232

- Overhead
- Right, short
- DIN 69880
- large adjustable ball-type nozzles



Longitudinal seat 353 6233

- Right
- large adjustable ball-type nozzles



Cover plate 353 6236

- Protection against contamination of the tool changer



Collet chuck ER 25 353 6237



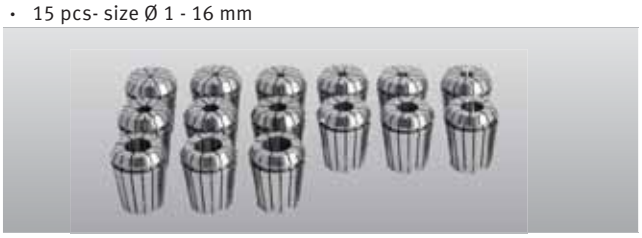
Key for collet chuck ER 25 353 6240



Collet chuck kit 344 1109

ER 25

- 15 pcs- size Ø 1 - 16 mm



Tool holder 353 6238

- pre-worked
- Blank round



Drill chuck 353 6239

- Clamping range 1 - 13 mm



Bohrstangenaufnahme Ø 10 mm	353 6241
Bohrstangenaufnahme Ø 12 mm	353 6242
Bohrstangenaufnahme Ø 16 mm	353 6243
Bohrstangenaufnahme Ø 20 mm	353 6244
Bohrstangenaufnahme Ø 25 mm	353 6245



Starter kit VDI 40

NEW

OPTIMUM® MASCHINEN - GERMANY

Starter kit	VDI 40
Item No	353 6116

Consisting of:

- 3 pcs Square transversal seat
- 1 pcs Square transversal seat overhead
- 1 pcs Longitudinal seat
- 5 pcs Bohrstangenaufnahme Ø 10 / 12 / 16 / 20 / 25 mm
- 3 pcs Cover plate
- 1 pcs Collet chuck ER 25
- 1 pcs Key for collet chuck ER 25
- 1 pcs Collet chuck kit ER 25
- 1 pcs Tool holder
- 1 pcs Bohrfutter

Square transversal seat 353 6251

- Right, short
- DIN 69880
- large adjustable ball-type nozzles



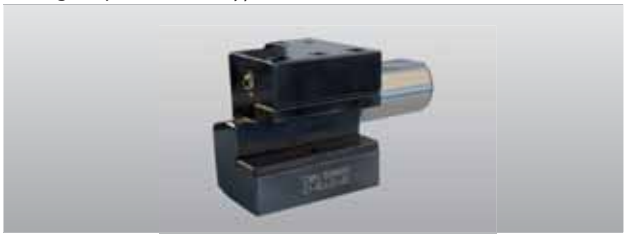
Square transversal seat 353 6252

- Overhead
- Right, short
- DIN 69880
- large adjustable ball-type nozzles



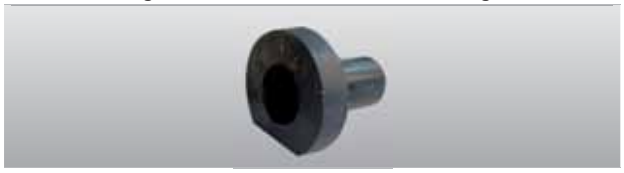
Longitudinal seat 353 6253

- Right
- large adjustable ball-type nozzles



Cover plate 353 6256

- Protection against contamination of the tool changer



Collet chuck ER 25 353 6257



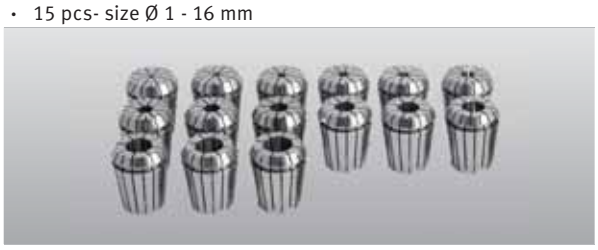
Key for collet chuck ER 25 353 6260



Collet chuck kit 344 1109

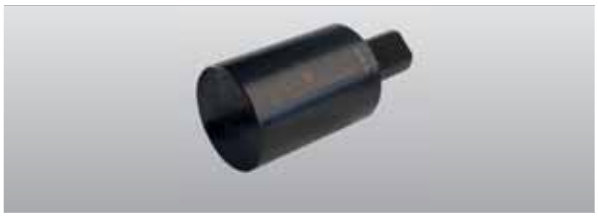
ER 25

- 15 pcs- size Ø 1 - 16 mm



Tool holder 353 6258

- pre-worked
- Blank round



Drill chuck 353 6259

- clamping range 1 - 13 mm



Boring bar shank Ø 10 mm	353 6261
Boring bar shank Ø 12 mm	353 6262
Boring bar shank Ø 16 mm	353 6263
Boring bar shank Ø 20 mm	353 6264
Boring bar shank Ø 25 mm	353 6265



Starter kit BT 30

Starter kit	BT 30
Item No	353 6107

Consisting of:
1 pcs Cone wiper
14 pcs Pull studs
3 pcs Collet chuck ER 32
1 pcs Key for collet chuck ER 32
2 pcs Collet chuck kit ER 32
1 pcs Drill chuck
2 pcs Weldon 6 mm
1 pcs Weldon 8 mm
1 pcs Weldon 10 mm
1 pcs Weldon 12 mm
1 pcs Weldon 16 mm
2 pcs Weldon 20 mm
1 pcs Adapter BT 30
1 pcs holder milling head
1 pcs Device for height adjustment
1 pcs Assembly and tool setting aid

Holder milling head 353 6306

- Seat 27 mm



Drill chuck 353 6303

- Clamping range 1 - 13 mm



Weldon holder 6 mm	353 6310
Weldon holder 8 mm	353 6311
Weldon holder 10 mm	353 6312
Weldon holder 12 mm	353 6313
Weldon holder 16 mm	353 6314
Weldon holder 20 mm	353 6315



Adapter 353 6305

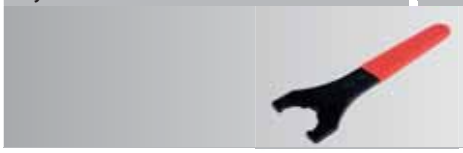
- BT 30 to MT 2



Collet chuck ER 32 353 6304



Key for collet chuck ER 32 353 6307



Collet chuck kit ER 32 344 1122

- 18 Collets Size Ø 1 - 16 mm



Device for height adjustment 353 6290

- Analogue type
- For rapid and easy determining of the reference point in the Z axis and/or to set the tools “zeroing” (e.g. for milling or drilling) without damaging the workpiece
- Housing height 50 mm



Assembly and tool setting aid 353 6193

- For easy and exact setting of the tools
- Vertical and horizontal seat for tools with taper shank



Cone wiper 353 6301



Pull studs 353 6302



Starter kit	BT 40 / Set 1
Item No	353 6105

Consisting of:

- 6 pcs Pull studs
- 2 pcs Collet chuck BT 40/ER32
- 1 pcs Collet chuck kit ER 32, 3 - 20 mm
- 1 pcs quick-action drill chuck0 - 13 mm
- 1 pcs Collet chuck BT 40 SLA 20-90
- 1 pcs adapter morse taper BT 40 - MT 3
- 1 pcs Surface milling cutter seat BT 40
- 1 pcs surface milling cutterr
- 1 pcs tool deposit
- 1 pcs holder milling head
- 1 pcs Device for height adjustment
- 1 pcs Assembly and tool setting aid
- 1 pcs Kit of parallel packing plates

Device for height adjustment 353 6180

- Analogue type
- Housing height 50 mm



Kit of parallel packing plates 353 6191

- Superfinished, accuracy of 0.01 mm
- Length 150 mm x Width 8.5 mm
- 18 pcs sizes 2 x 14 mm/2 x 16 mm/2 x 20 mm/2 x 24 mm/ 2 x 30 mm/2 x 32 mm/2 x 36 mm/2 x 40 mm/2 x 44 mm



Assembly and tool setting aid BT 40 353 6187

- For easy and exact setting of the tools
- Vertical seat for tools with taper shank
- Weight 13 kg



Milling chuck - BT 40 SLA 20 - 90 353 6186

- To seat the 3-D push button



Quick-action drill chuck 0 - 13 mm - BT 40 353 6183

- Including hook spanner for closing



Collet chuck BT 40/ER 32 353 6182

- Accuracy 0,005 mm



Collet chuck kit ER 32 344 1122

- **18 collets 3 - 20 mm**



Adapter Morse taper BT 40 - MT 3 353 6184



Surface milling cutter seat - BT 40 (27 mm) 353 6190



Surface milling cutter 353 6189

- 63 mm · Hole 27 mm · Including cutting inserts

Spare cutting inserts 353 0196

- 10 pcs



Pull studs BT 40 - 40 x 45°	353 6185
Pull studs Sk 40 M16	353 6179
Pull studs BT 40 HC 40	353 6178

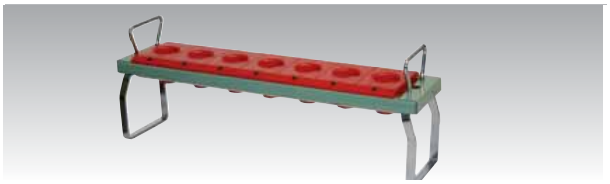
- for F210 TC assembly tools



Fig. BT Fig. SK

Tool deposit for BT 40 tools 353 6188

- Dimensions L x W x H: 595 x 143 x 225 mm



Starter kit BT 40/Set 2

Starter kit	BT 40 / Set 2
Item No	353 6108

Consisting of:

- 1 pcs Cone wiper
- 15 pcs Pull studs
- 3 pcs Collet chuck ER 32
- 1 pcs Key for collet chuck ER 32
- 2 pcs Collet chuck kit ER 32, 1 - 16 mm
- 1 pcs Drill chuck 1 - 13 mm
- 2 pcs Weldon 6 mm
- 1 pcs Weldon 8 mm
- 1 pcs Weldon 10 mm
- 1 pcs Weldon 12 mm
- 1 pcs Weldon 16 mm
- 2 pcs Weldon 20 mm
- 1 pcs Adapter BT 40 to MT 3
- 1 pcs holder milling head mit Seat 27 mm
- 1 pcs Device for height adjustment
- 1 pcs Assembly and tool setting aid

Holder milling head 353 6336

- Seat 27 mm



Drill chuck 353 6333

- Clamping range von 1 - 13 mm



Weldon holder 6 mm	353 6340
Weldon holder 8 mm	353 6341
Weldon holder 10 mm	353 6342
Weldon holder 12 mm	353 6343
Weldon holder 16 mm	353 6344
Weldon holder 20 mm	353 6345



Adapter 353 6335

- BT 40 to MT 3



Cone wiper 353 6331



Collet chuck ER 32 353 6334



Key for collet chuck ER 32 353 6307



Collet chuck kit ER 32 344 1122

- 18 Collets Size Ø 1-16 mm



Device for height adjustment 353 6290

- Analogue type
- For rapid and easy determining of the reference point in the Z axis
- and/or to set the tools “zeroing” (e.g. for milling or drilling) without
- damaging the workpiece
- Housing height 50 mm



Assembly and tool setting aid 353 6194

- For easy and exact setting of the tools
- Vertical and horizontal seat for tools with taper shank



Pull studs 353 6332



Starter kit SK40/DIN 69871

NEW

OPTIMUM® MASCHINEN - GERMANY

Starter kit	SK 40 / DIN 69871
Item No	353 6109

Consisting of:
1 pcs cone wiper
6 pcs Pull studs
3 pcs Collet chuck ER 32
1 pcs Key for collet chuck ER 32
2 pcs Collet chuck kit ER 32, 1 - 16 mm
1 pcs Drill chuck 1 - 13 mm
2 pcs Weldon 6 mm
1 pcs Weldon 8 mm
1 pcs Weldon 10 mm
1 pcs Weldon 12 mm
1 pcs Weldon 16 mm
2 pcs Weldon 20 mm
1 pcs Adapter SK 40 to MT 3
1 pcs holder milling head mit Seat 27 mm
1 pcs Device for height adjustment
1 pcs Assembly and tool setting aid

Holder milling head 353 6366

- Seat 27 mm



Drill chuck 353 6363

- Clamping range von 1 - 13 mm



Weldon holder 6 mm	353 6370
Weldon holder 8 mm	353 6371
Weldon holder 10 mm	353 6372
Weldon holder 12 mm	353 6373
Weldon holder 16 mm	353 6374
Weldon holder 20 mm	353 6375



Adapter 353 6365

- SK 40 to MT 3



Cone wiper 353 6331



Collet chuck ER 32 353 6364



Key for collet chuck ER 32 353 6307



Collet chuck kit ER 32 344 1122

- 18 collets size Ø 1-16 mm



Device for height adjustment 353 6290

- Analogue type
- For rapid and easy determining of the reference point in the Z axis
- and/or to set the tools “zeroing” (e.g. for milling or drilling) without
- damaging the workpiece
- Housing height 50 mm



Assembly and tool setting aid 353 6193

- For easy and exact setting of the tools
- Vertical and horizontal seat for tools with taper shank



Pull studs 353 6362

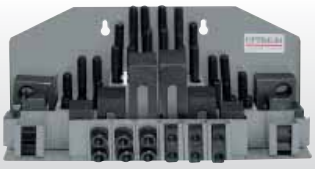
- 15 pcs



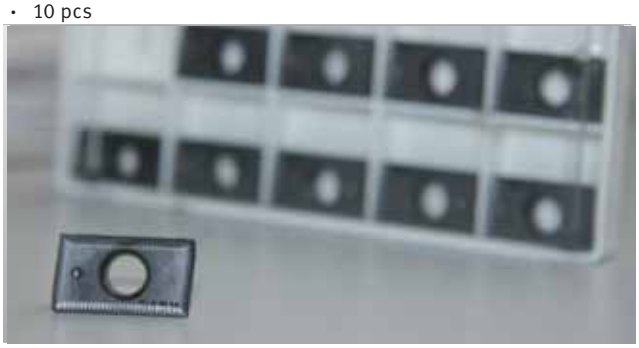
Surface milling cutter without cuttin inserts	
Surface milling cutter Ø 63 mm hole, 27 mm	353 6390
Surface milling cutter Ø 50 mm hole, 22 mm	353 6391



Clamping tool kit SPW 58 pcs	
• Metric threads	
• 24 pcs draw-in bolts	
• 6 pcs T-slots	
• 6 pcs nuts	
• 4 pcs elongation nuts	
• 6 pcs step blocks	
• 12 pcs clamping blocks	
• Practical wall bracket	
SPW 8	335 2015
• T-slots 10 mm; draw bar thread M 8	
SPW 10	335 2016
• T-slots 12 mm; draw bar thread M 10	
SPW 12	335 2017
• T-slots 14 mm; draw bar thread M 12	
SPW 14	335 2018
• T-slots 16 mm; draw bar thread M 14	
SPW 16	335 2019
• T-slots 18 mm; draw bar thread M 16	

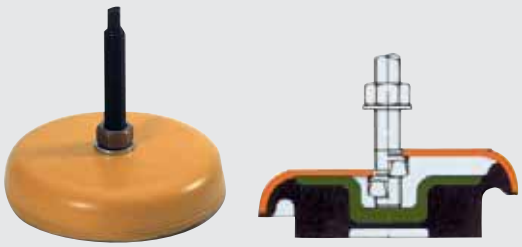


cutting inserts	353 6392
• APKT 1604 PDTR-ME14 C15M	
• 10 pcs	



Vibration damping machine foot	Item No
• Machines and devices can be installed without anchoring and exactly levelled per height adjustment	
• Machine capability is improved by efficient impact and vibration damping	
SE 1	338 1012
SE 2	338 1016
SE 3	338 1018

Charge	SE 1	SE 2	SE 3
Milling machines	340 Kg	460 kg	1'600 kg
Sawing/ general machines	570 kg	1'460 kg	3'500 kg
Thread	M12	M 16	M 20
Base Ø/Base height	120/32	160/35	185/39



CNC-instruction - Information

CNC-instruction indoor	
Item No.	900 0509
on-site operation	
excl. travel expenses / overnight stay	

Accounting unit: 1 day
The commissioning includes the alignment, the acceptance and the functional test of the machine. This applies under the condition that the machine is unpacked, that exterior corrosion protection is removed, that the machine is cleaned, moved to the installation place and connected to the electrical or pneumatic energy supply by the customer.
Subsequently your employees are briefly instructed in the function, operation, preparation, diagnostic and maintenance of the machine.
Duration of the commissioning and instruction: 8 hours/day, including the time for arrival and departure plus travelling cost, plus accommodation at cost, if applicable.
The same applies for deviating and additional services which are separately ordered on site; such services will be invoiced on a time and material basis.
If it is necessary and desired to install and connect the machine, the customer engages himself to support our assemblers using his best efforts and to make available the necessary assistants, tools and appliances at his own cost.

Machine vices for standard and individual manufacturing on CNC milling machines and machining centres

Hydraulic vice HCV

Modular - machine vice of high accuracy and repeatability for standard and individual manufacturing on CNC milling machines and machining centres
Solid structure for milling
Hardened and ground guide surfaces
Booster system, little effort, high pressure output when clamping
Large clamping range due to locking bolts with different hole spacings
Manufactured of high-quality stainless steel
Long service life due to high quality type
High clamping power

HCV 105	353 6210
HCV 125	353 6214
Non-rigid jaws 2 pcs for HCV125	353 6221
L-jaws 2 pcs for HCV125	353 6222

Technical data		L	B	C	j	a	T-slots	Clamping pressure	Clamping force	kg
HCV 105	mm	442	107	170	42	107	16	24.5 KN	2500 kg	22
HCV 125	mm	478.5	140	205	45	126	18	40 KN	4000 kg	35.4



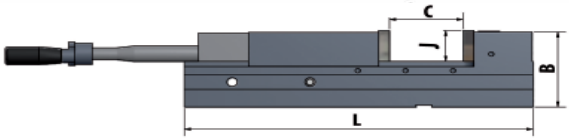



Fig.: HCV 105


Fig.: HCV 125


Fig.: Non-rigid jaws


Fig.: L-jaws

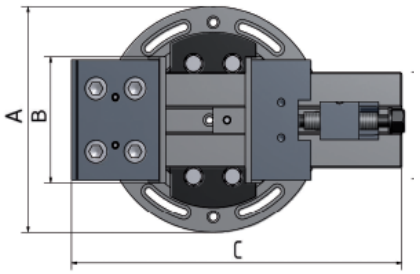
OPTIMUM precision modular vices PNM

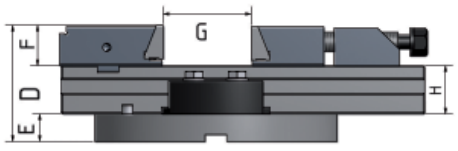
Modular vice for standard and individual manufacturing on CNC milling machines and machining centres
Rotary table
Particularly suitable for tool manufacturing, mould making and construction of jigs and fixtures
Material made of alloyed tool steel
Hardened and ground guideways
Rapid alignment over longitudinal and transversal grooves
High precision
High clamping force
Horizontally and vertically applicable
Little installation height
Easy handling

PNM 100	335 5551
PNM 125	335 5553

Technical data		A	B	C	D	E	F	G	H	kg
PNM 100	mm	180	100	270	85	20	30	95	35	10.3
PNM 125	mm	226	125	345	103	23	40	150	40	18.2


Fig.: PNM 100





Please note

General notes regarding the operation of our machines:

- Generally, our machines have to be operated only under supervision. However, if you leave the machine during operation you are acting grossly negligent.
- Please find the indications regarding the accuracy of the machines in the technical data on the catalogue pages. If you cannot find the corresponding values there, please do not hesitate to contact info@ optimum-maschinen. de in order to obtain more detailed information.
- The indicated accuracies are achieved under standard conditions (proper installation of the machine and environmental temperatures

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MASCHINEN - GERMANY

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Metal working machines for craftsmen and industry

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The entry-level models



The installation professionals



The solid compressors for craftsmen



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Screw-type compressors and stationarscrew compressors



Compressed air distribution Pipe plug systems

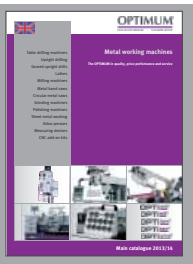


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MASCHINEN - GERMANY

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- Technical and optical modifications are reserved.
- Illustrations partially including optional accessories
- The delivery is performed exclusively according to our terms of delivery and payment.
- The machines are delivered partially disassembled for transport-technical reasons.
- The products are sold by specialized trade.
- The products remain our property until they are completely paid.
- Our ownership reservation shall remain valid upon third-party resale.
- In the commercial sector, the legal terms of warranty shall apply.
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